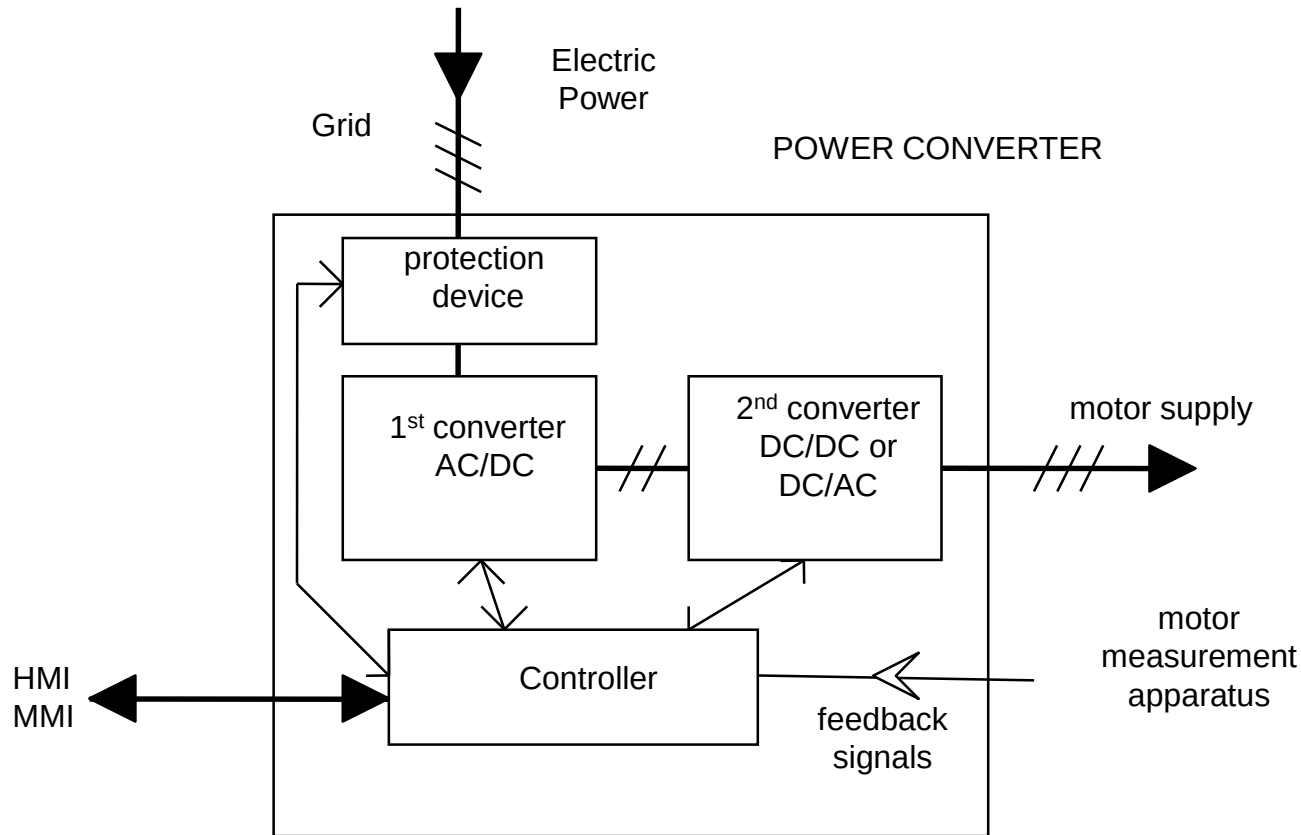
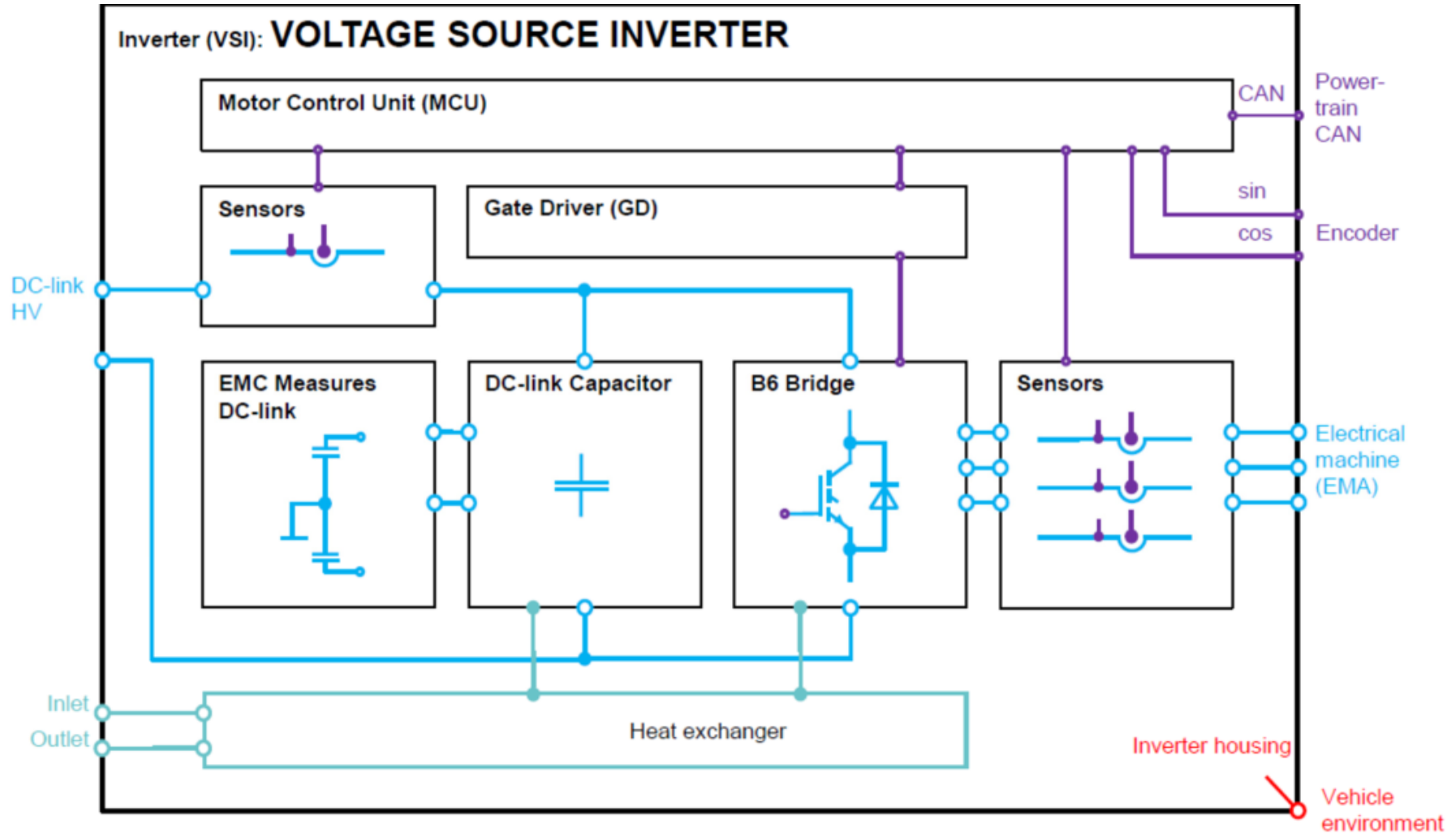


AC/DC conversion system

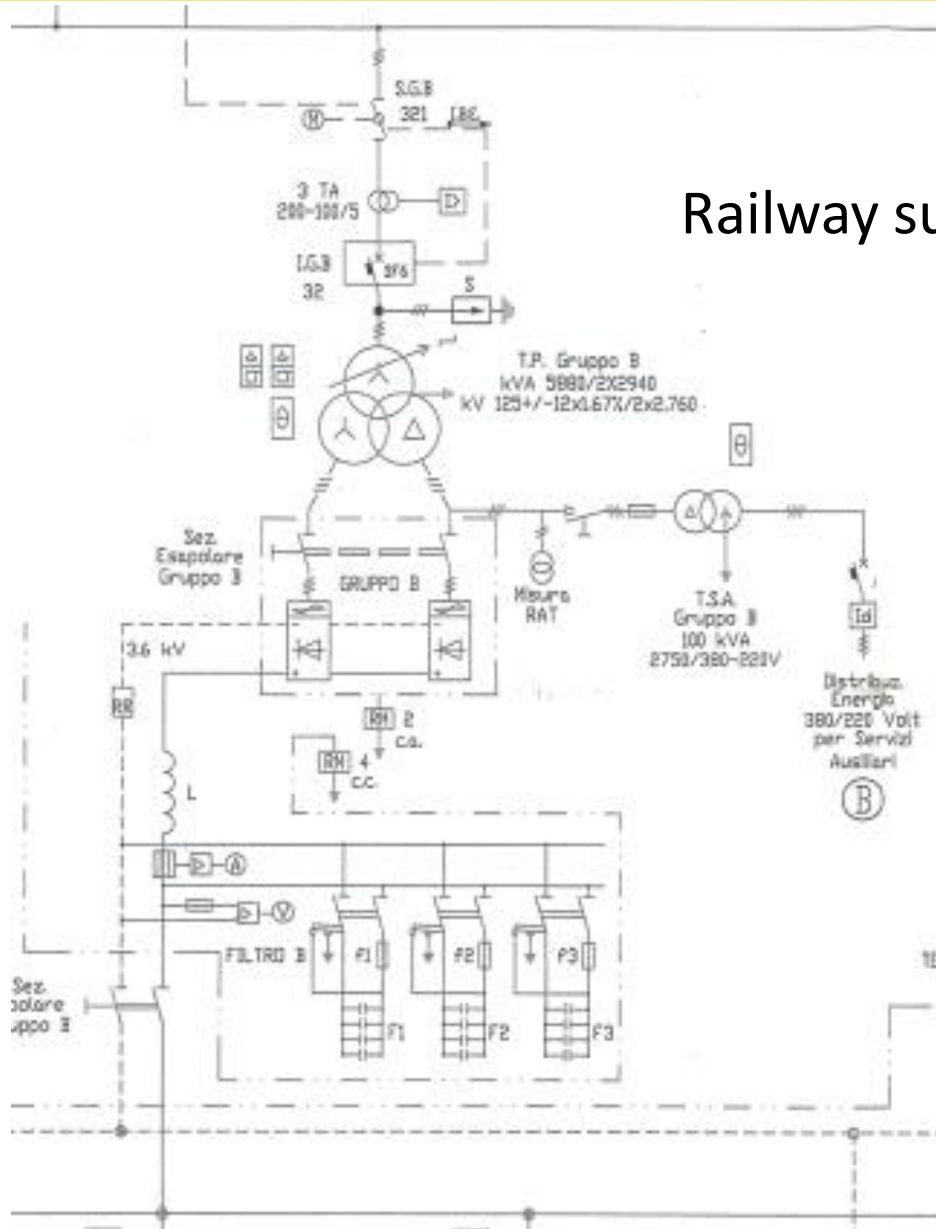


- Charging stations
- AC 25kV Trains
- Railway substations

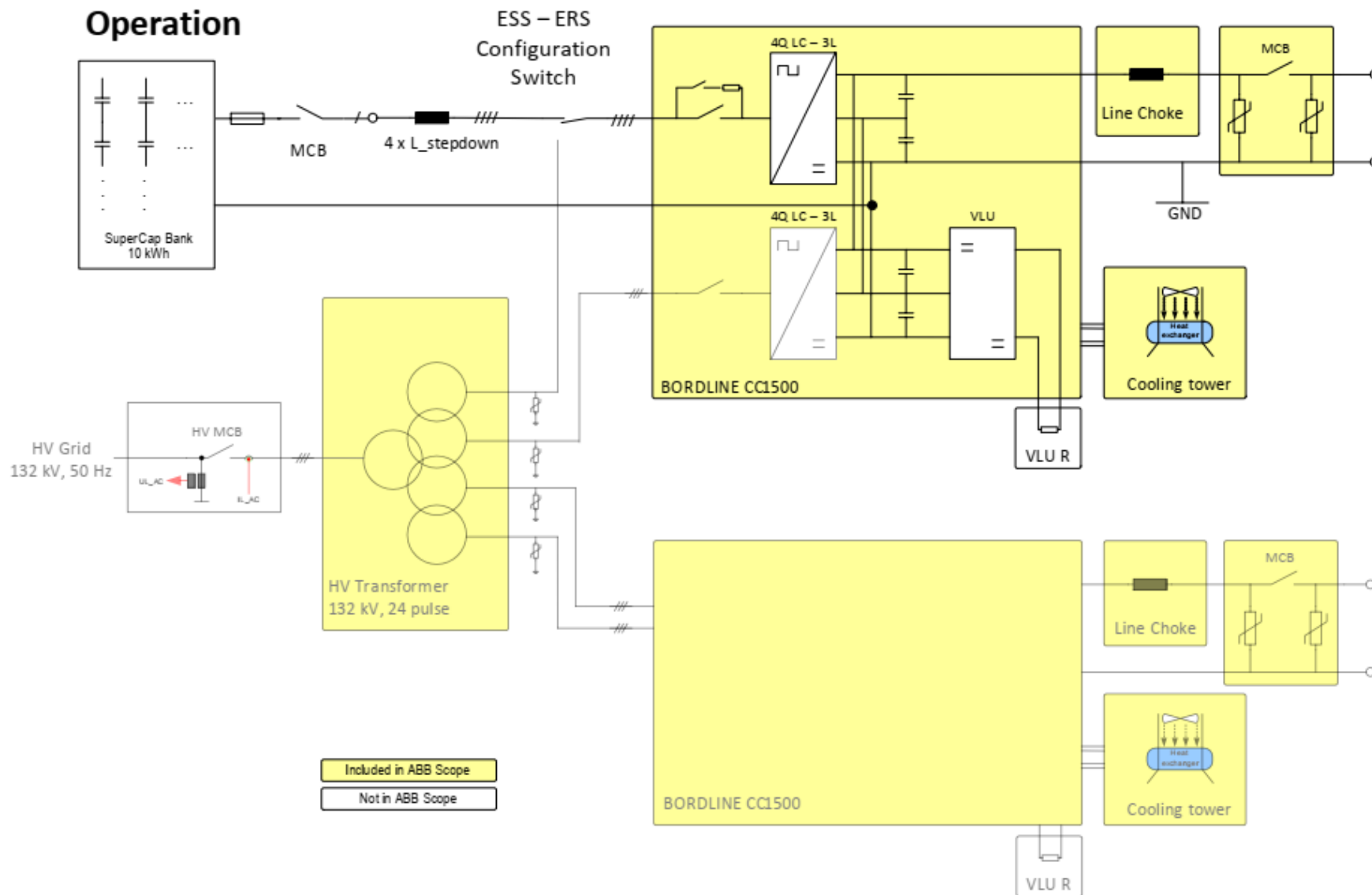
EV power converter



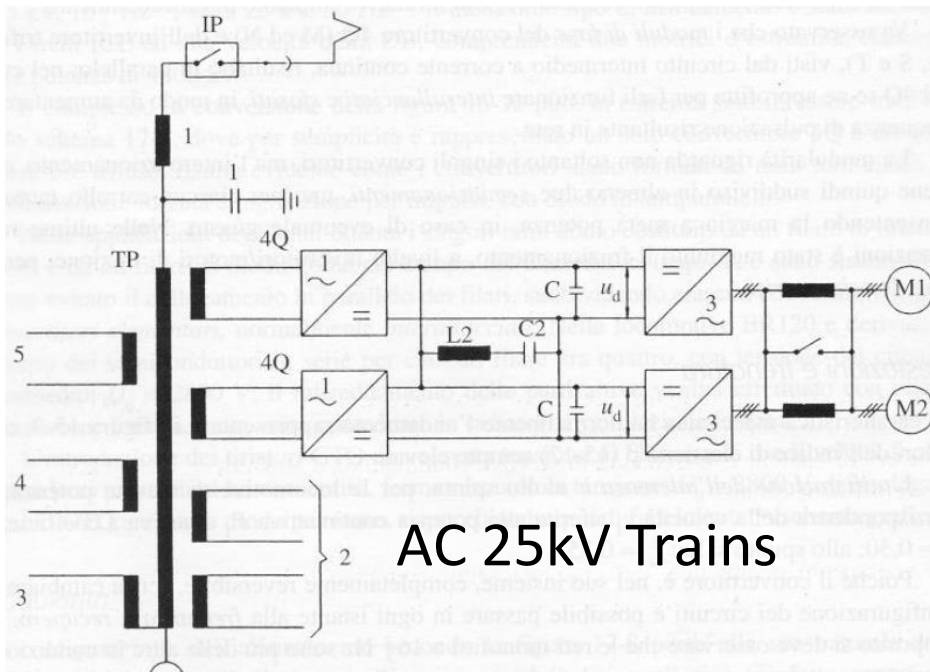
Railway substations (Forlì)



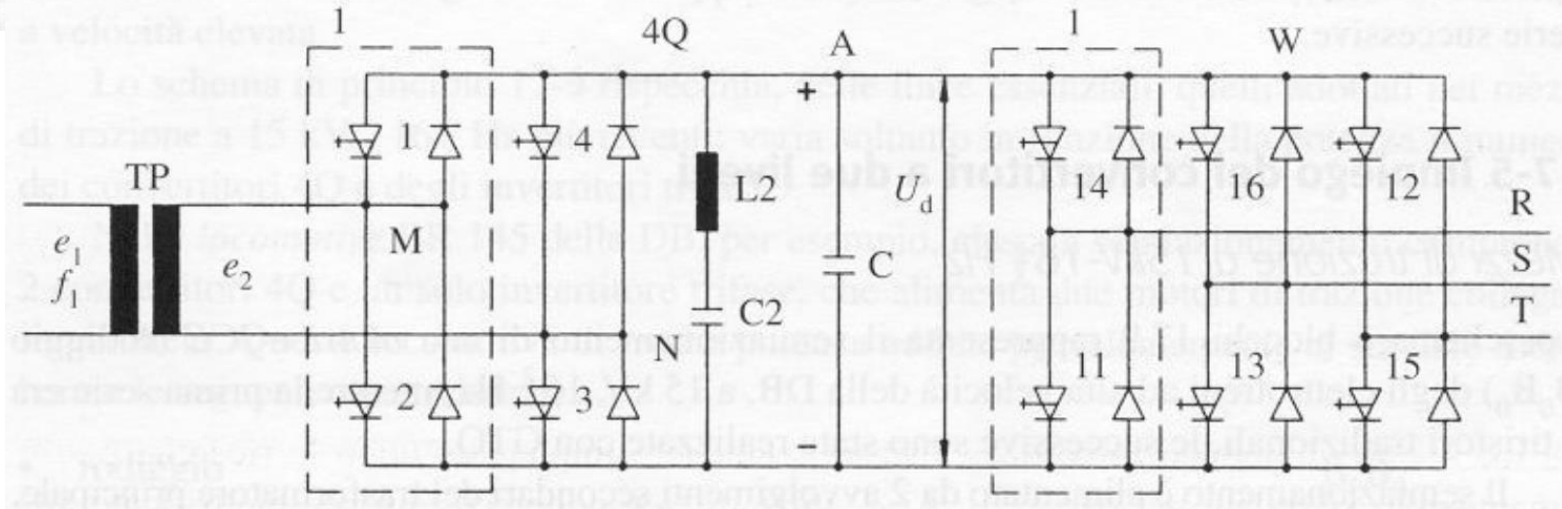
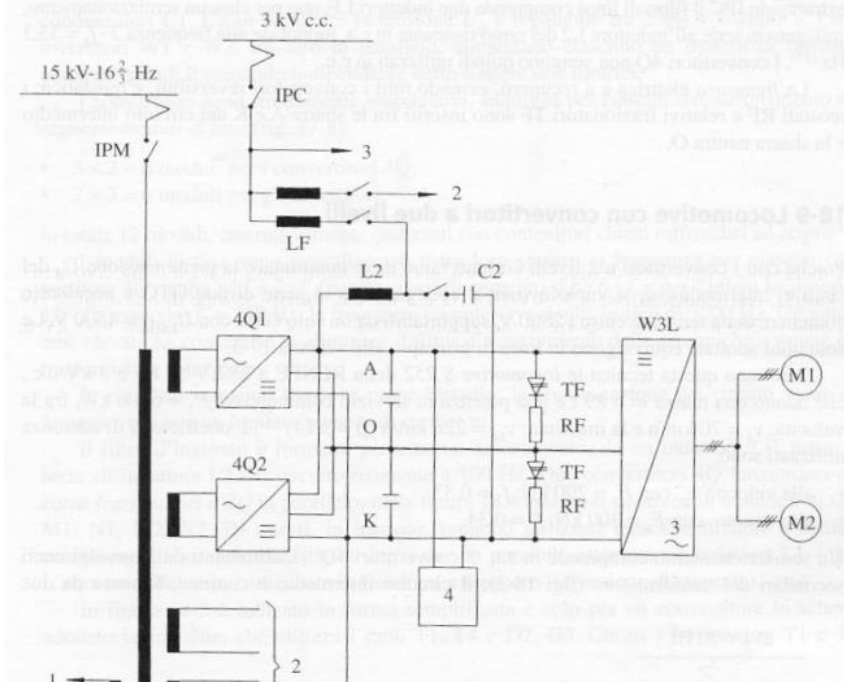
ESS in Operation



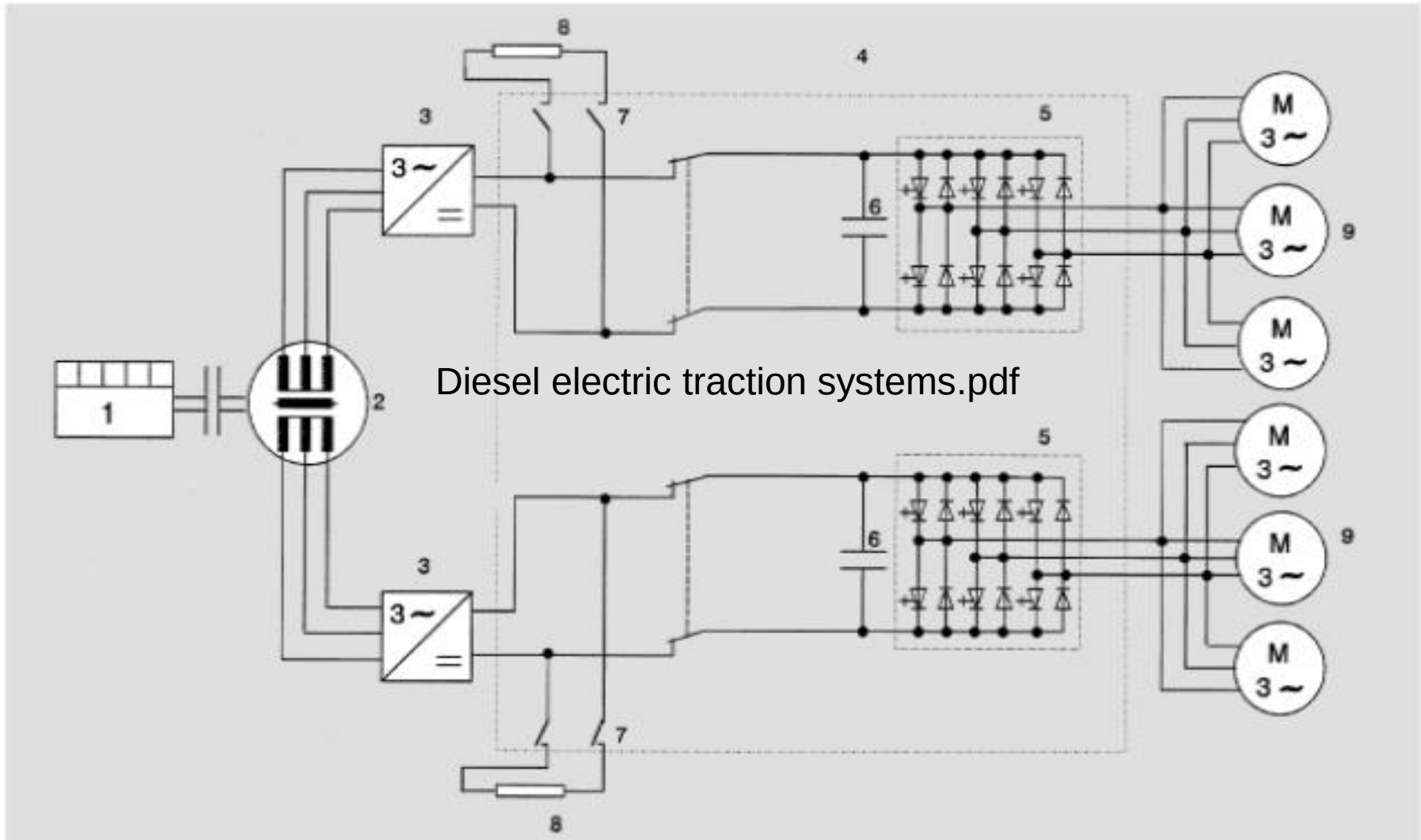
Four-quadrant rectifier



AC 25kV Trains



Diesel-electric traction systems (train, ship)



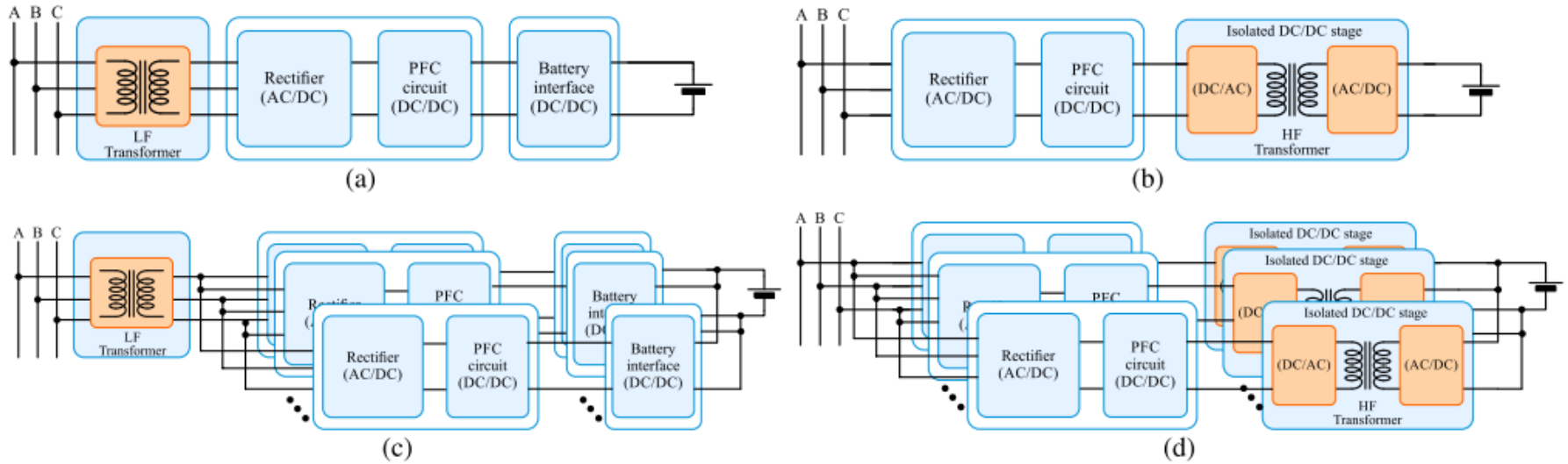


Fig. 1. Simplified block diagram of conventional dc fast charger power conversion systems. (a) Single-module charger with a nonisolated dc/dc converter. (b) Single-module charger with an isolated dc/dc converter. (c) Multiple paralleled modules shown in (a). (d) Multiple paralleled modules shown in (b).

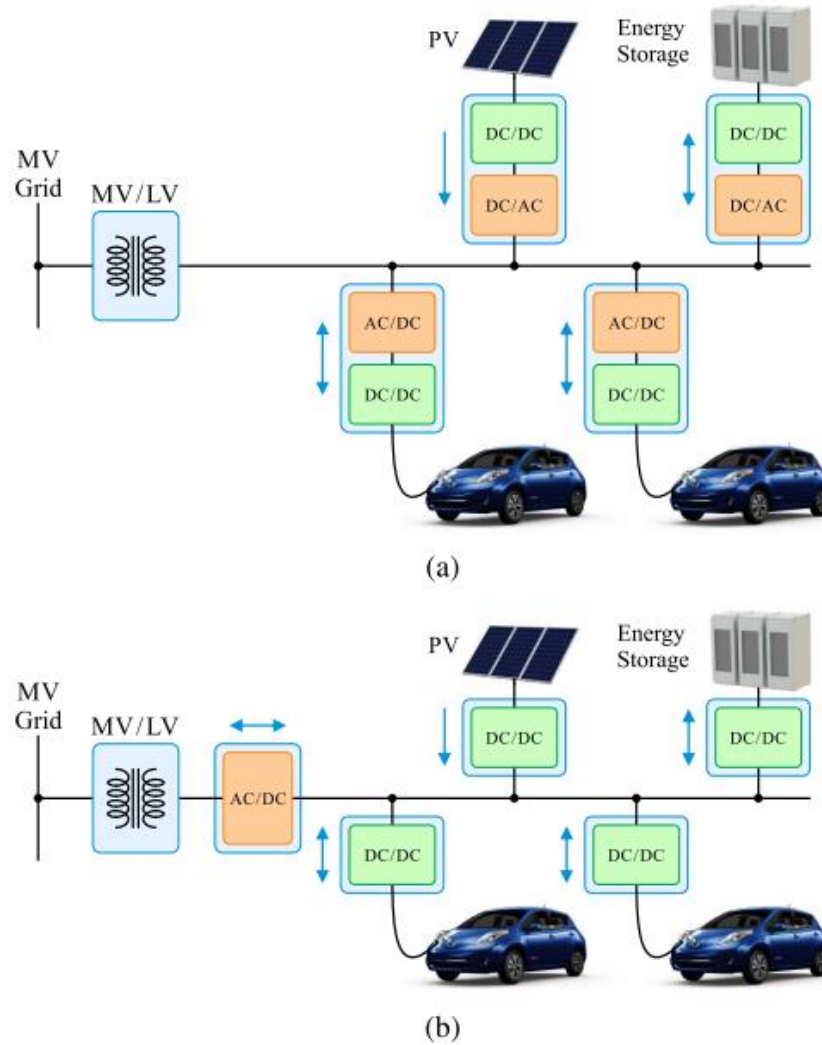
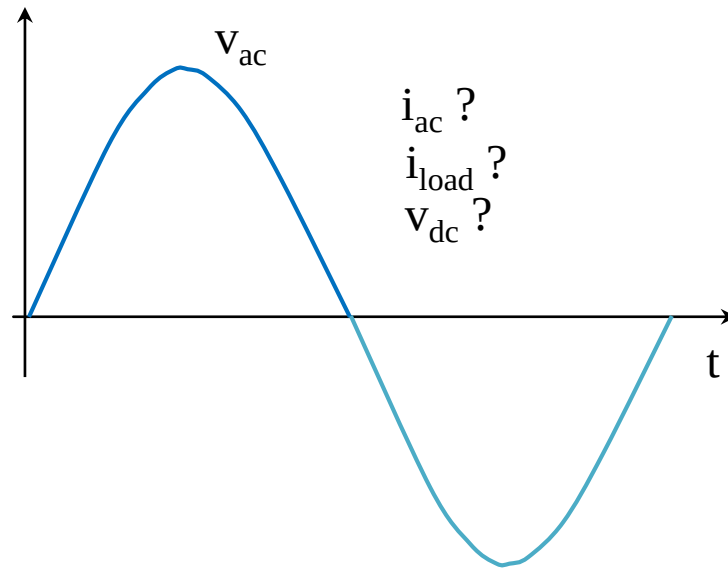
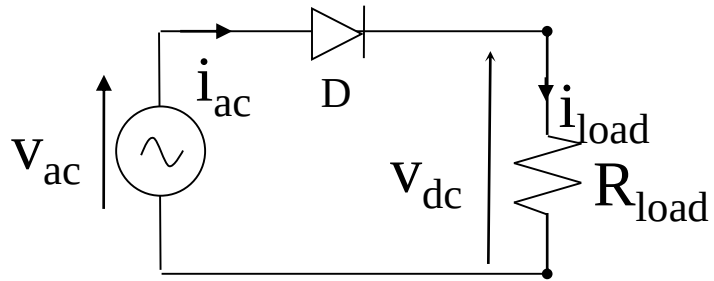
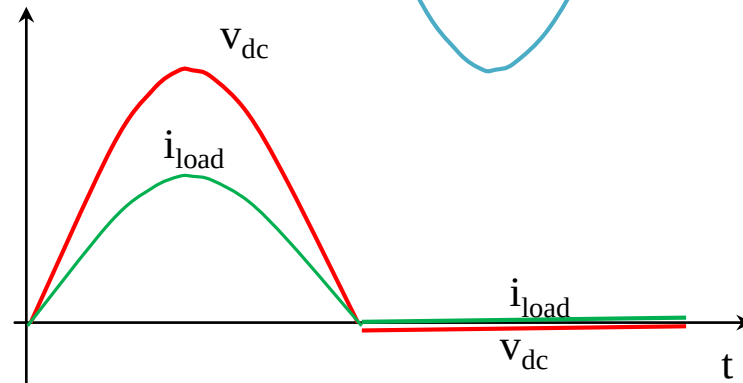
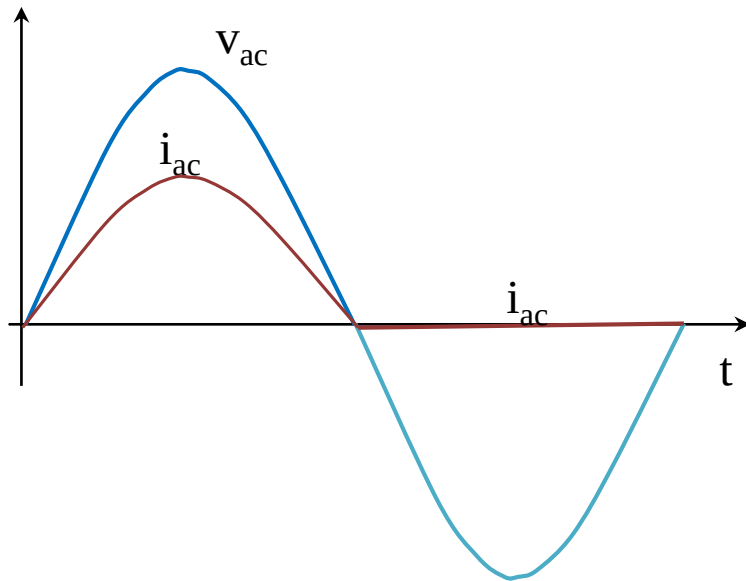
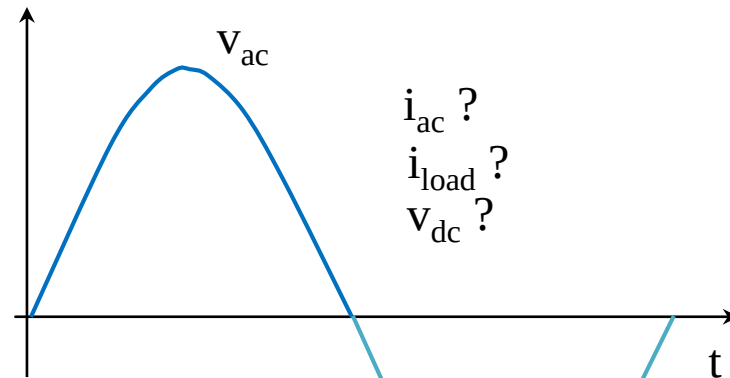
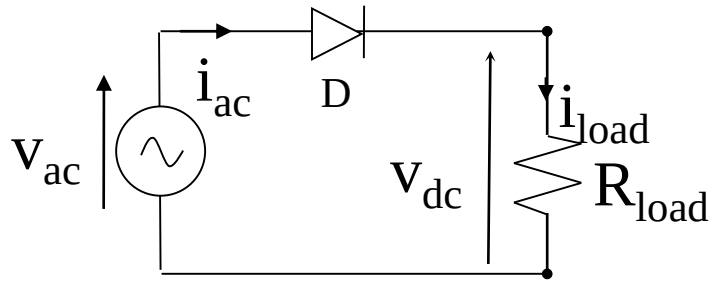


Fig. 3. Configurations for XFC stations. (a) AC-connected system. (b) DC-connected system.

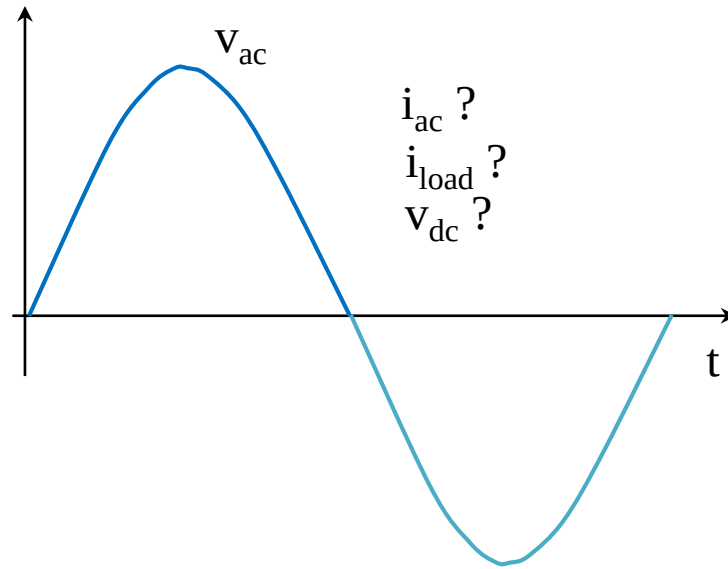
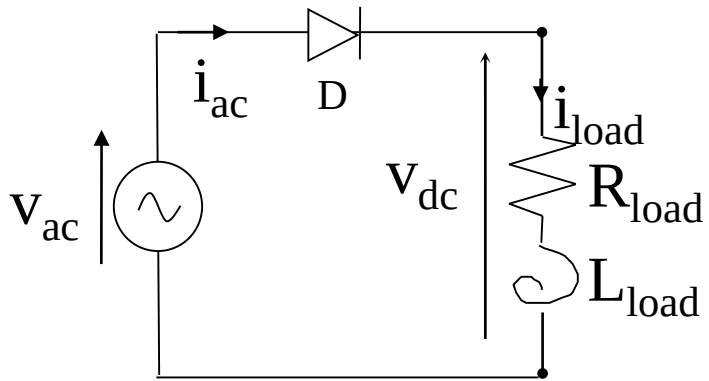
AC/DC: Test



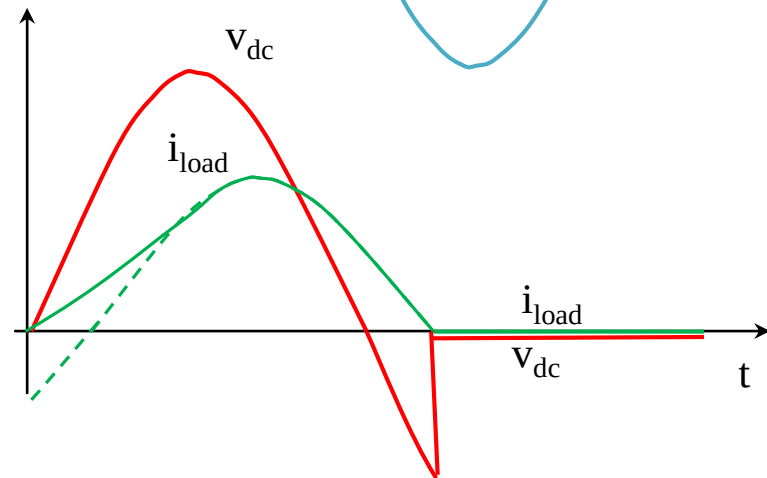
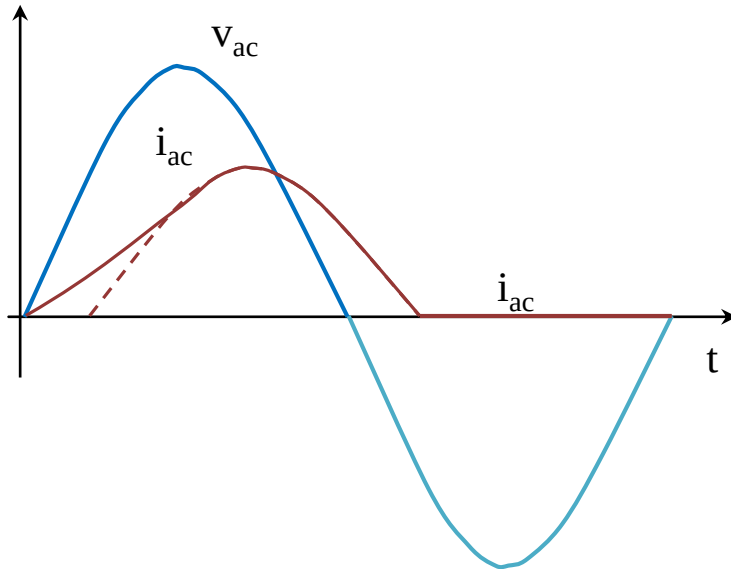
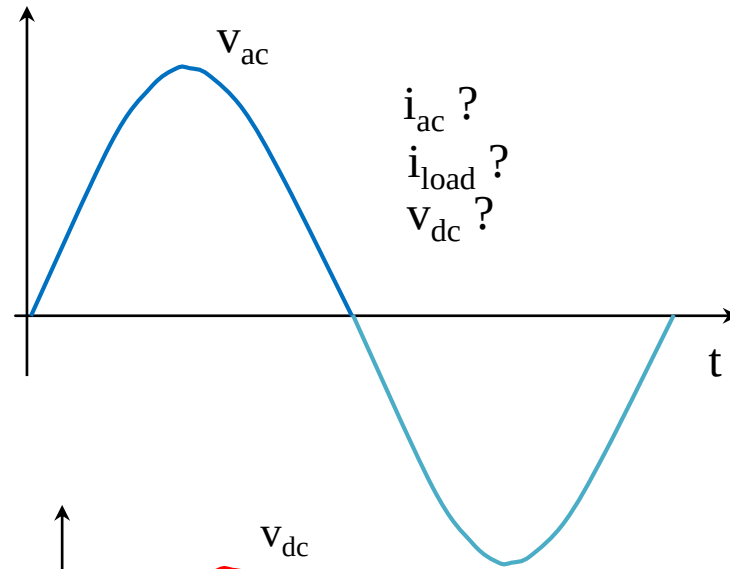
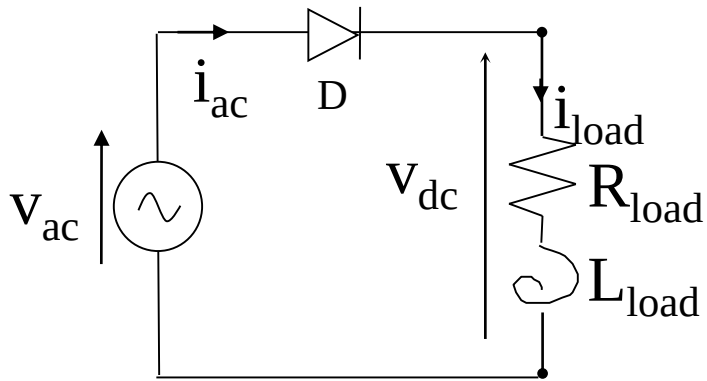
AC/DC: Test



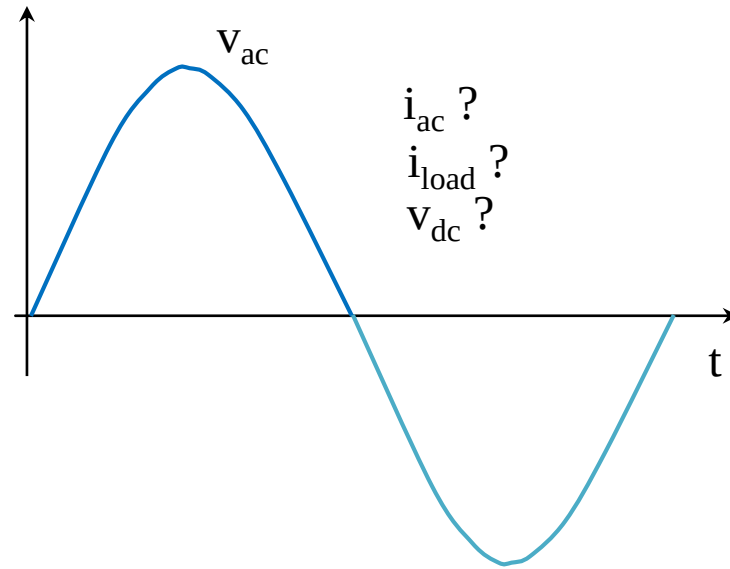
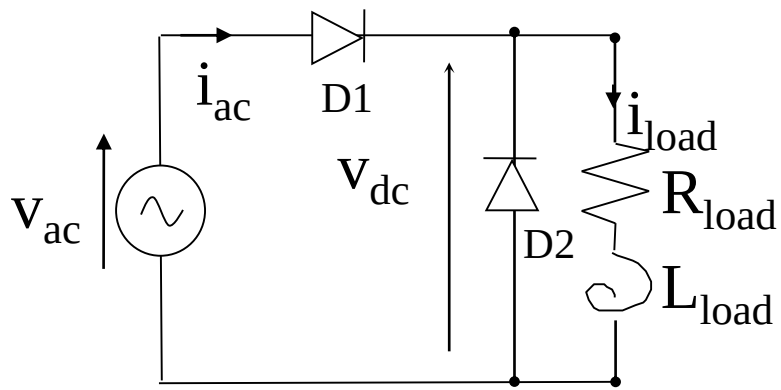
AC/DC: Test



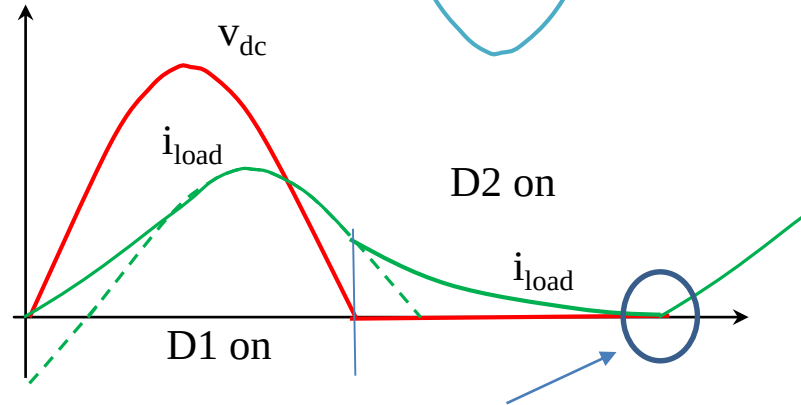
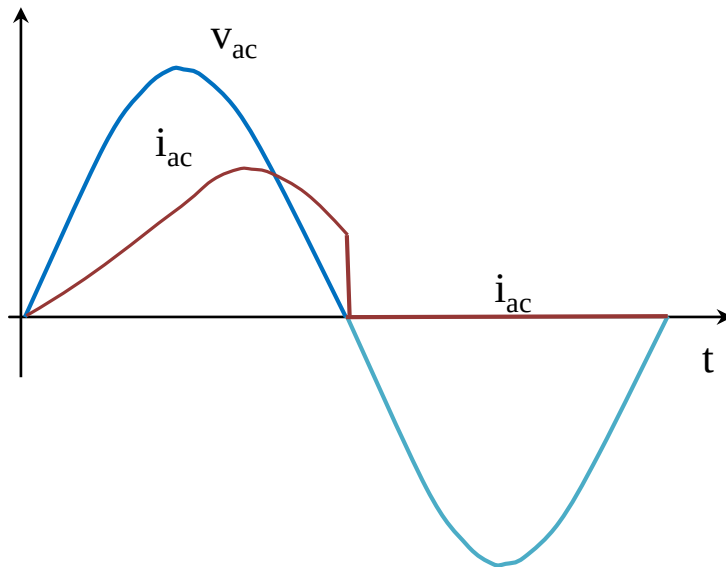
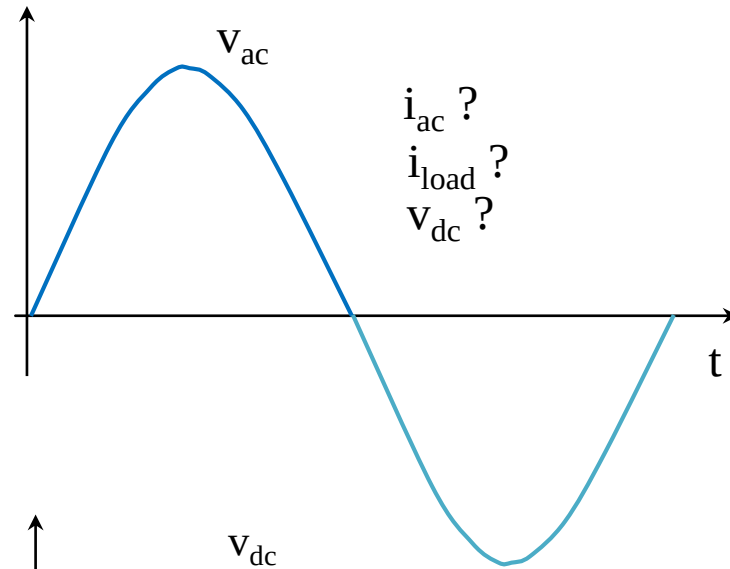
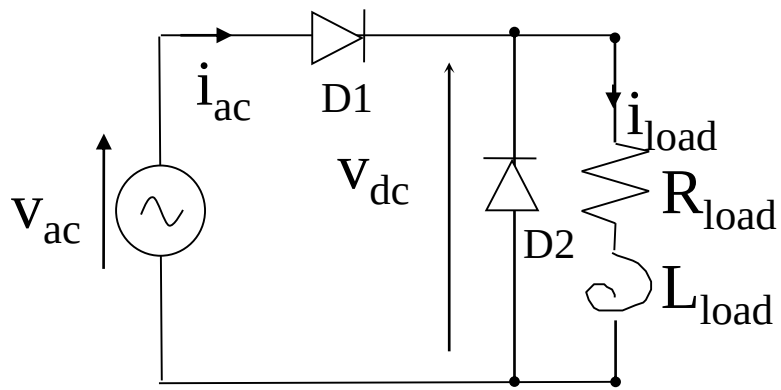
AC/DC: Test



AC/DC: Test

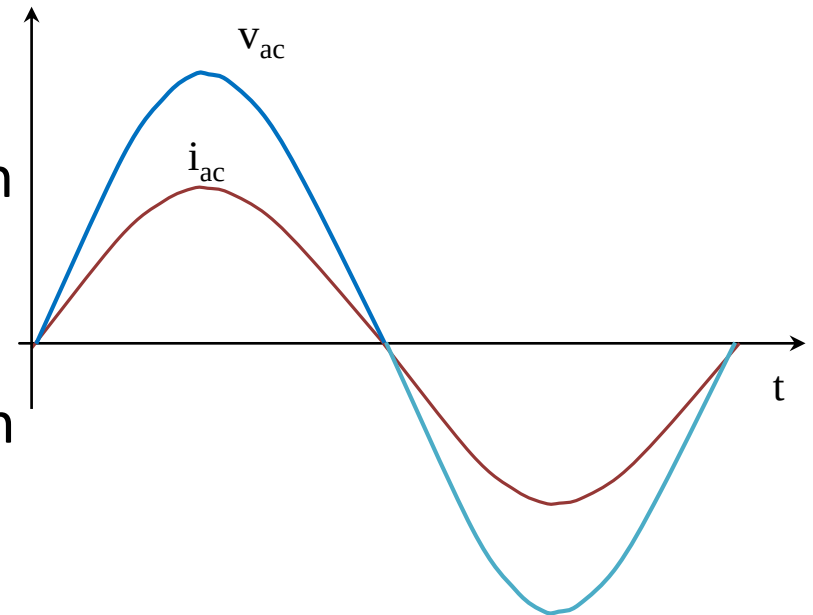
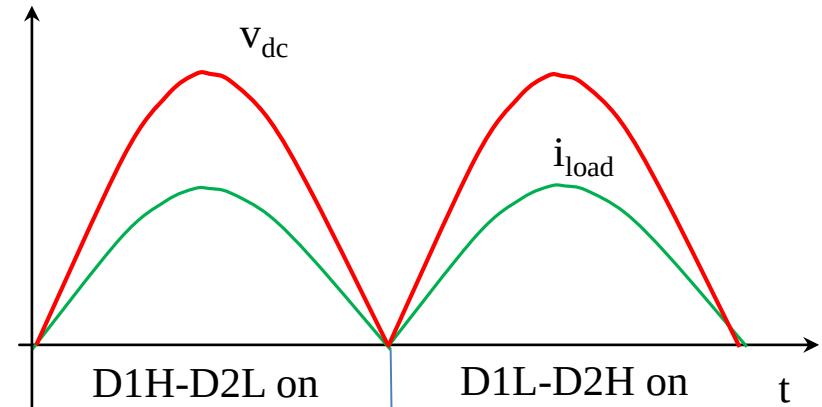
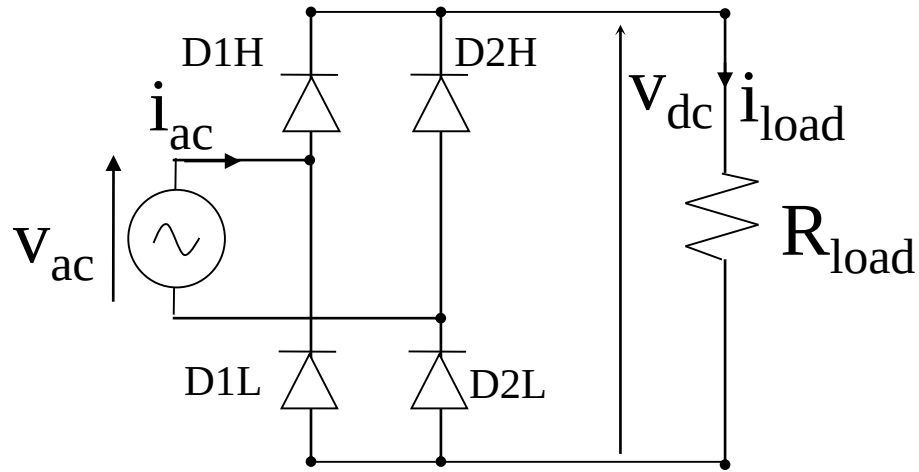


AC/DC: Test



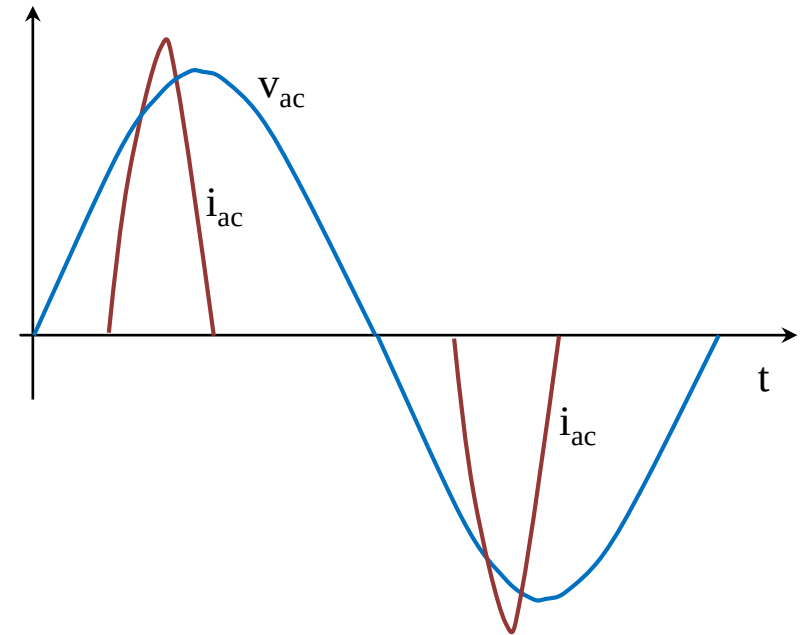
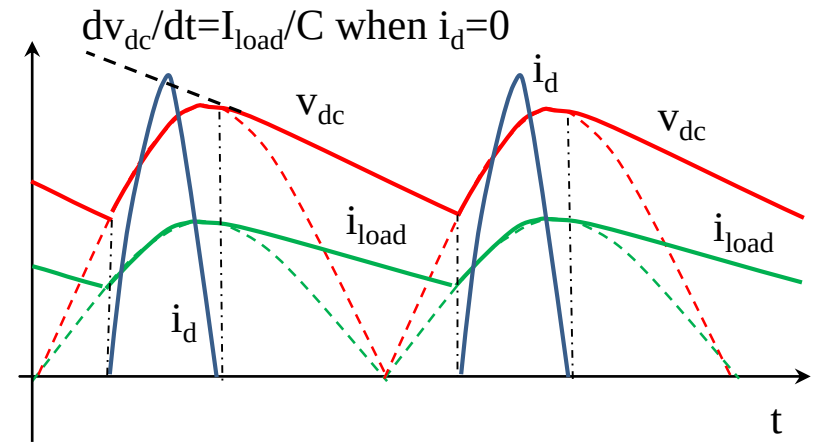
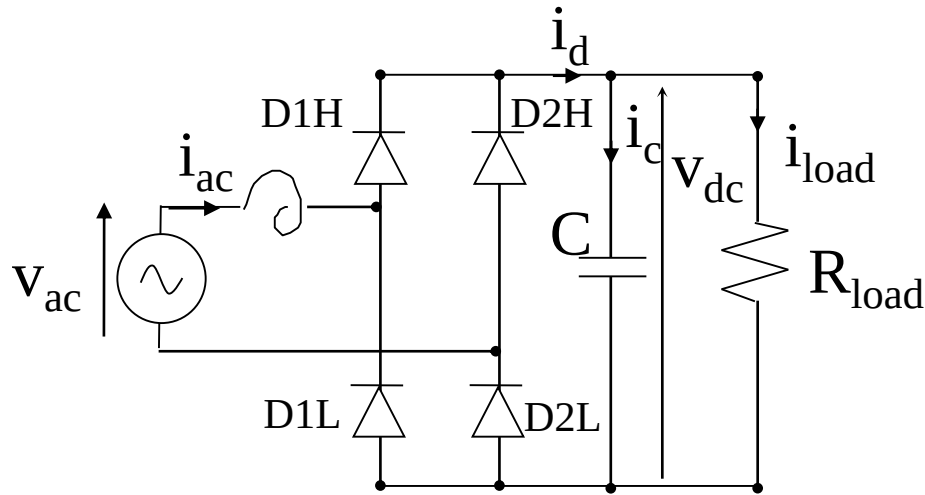
Continuous or Discontinuous mode?

Diode bridge



- Between N diodes with a common cathode, it conducts the one with the highest anode voltage
- Between N diodes with a common anode, it conducts the one with the lowest cathode voltage

Diode bridge with a dc-bus capacitor

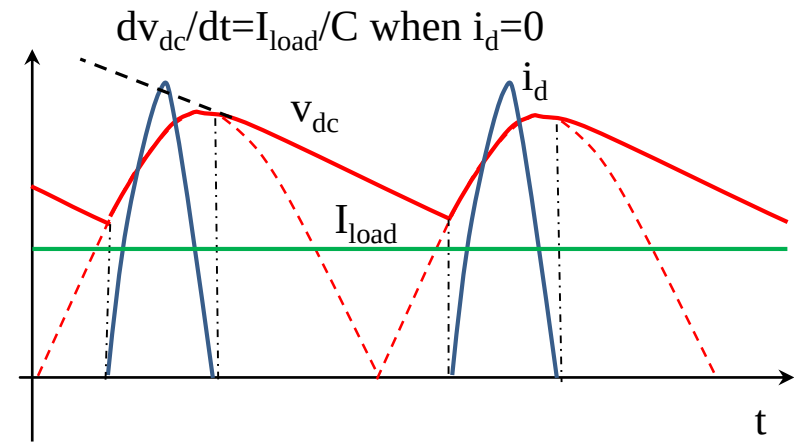
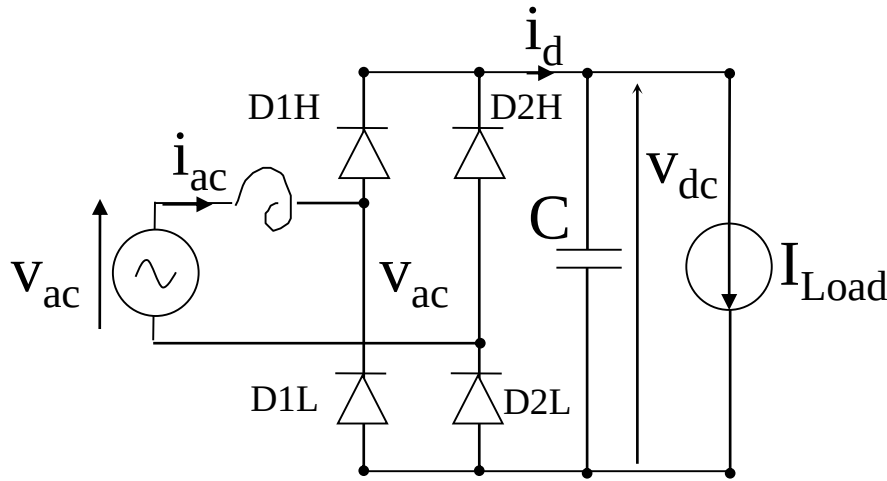


The diode bridge is disconnected from the load when i_d tries to become negative:

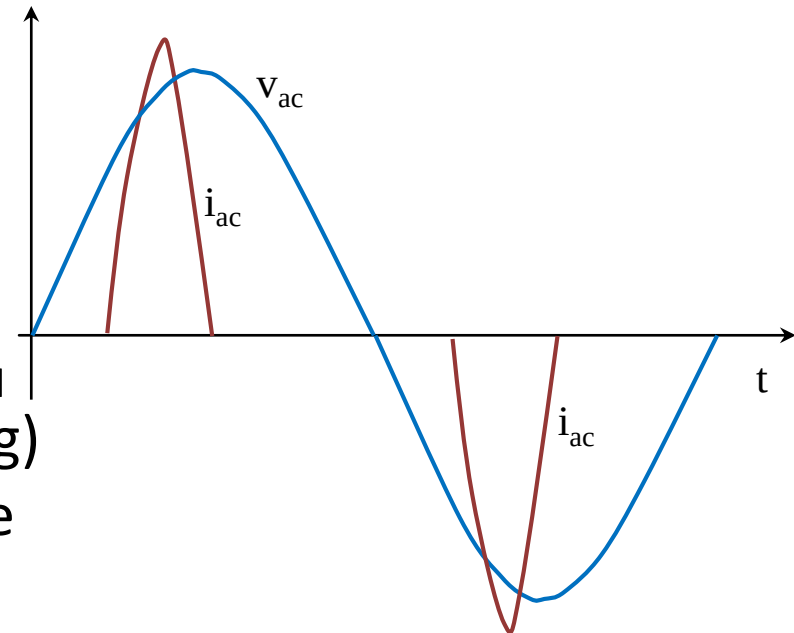
$$i_d = i_c + i_{load} = C \frac{dv_{dc}}{dt} + i_{load}$$

The period during which the grid is connected to the load is less than a semiperiod of the ac voltage

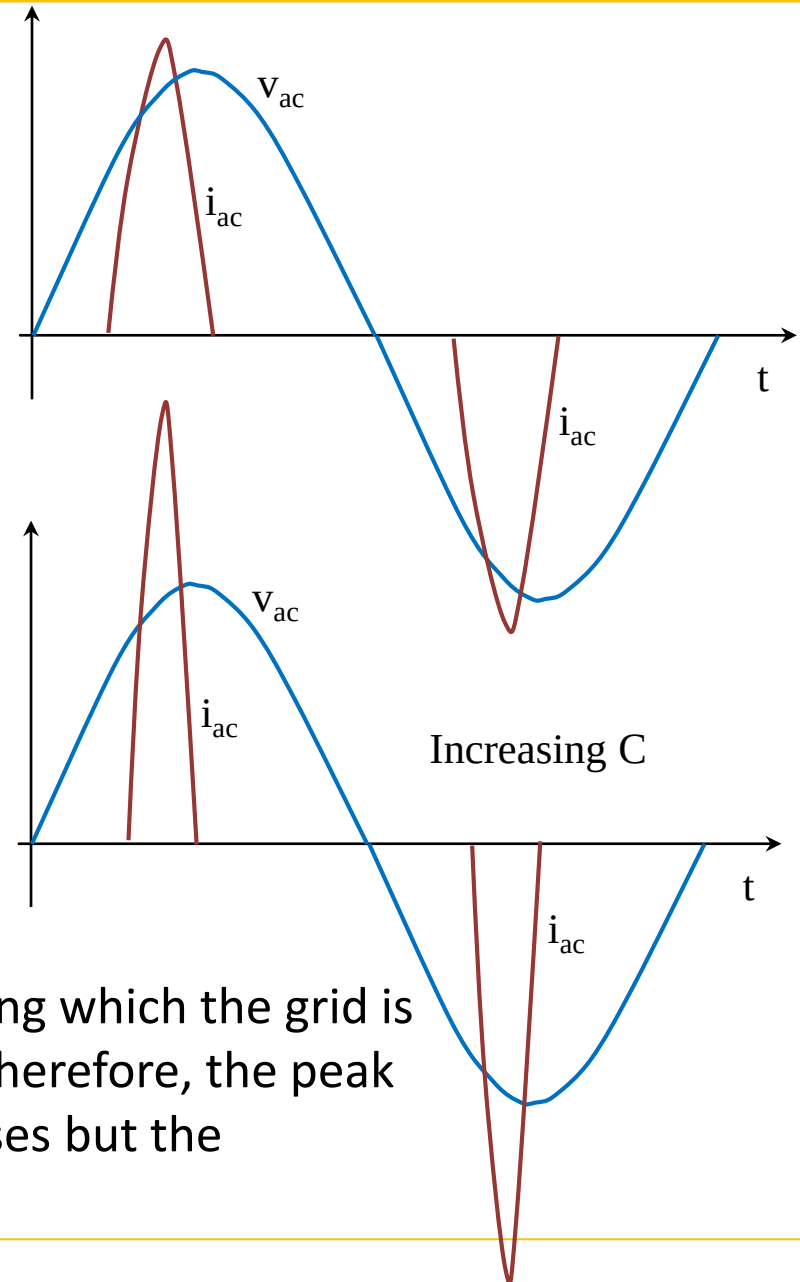
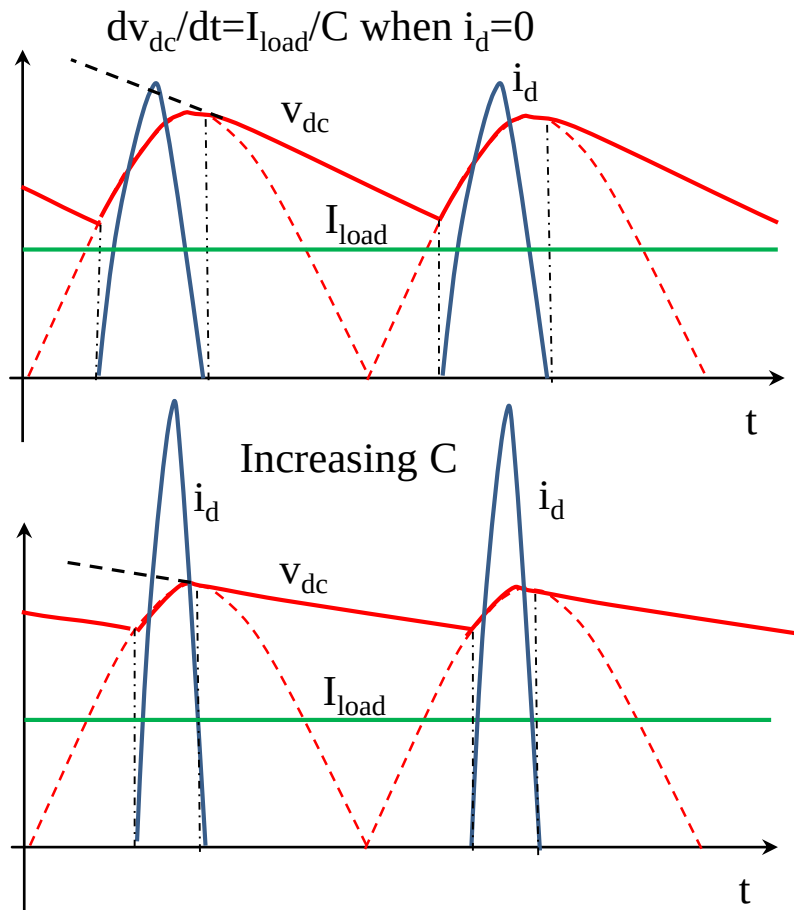
Diode bridge with a dc-bus capacitor



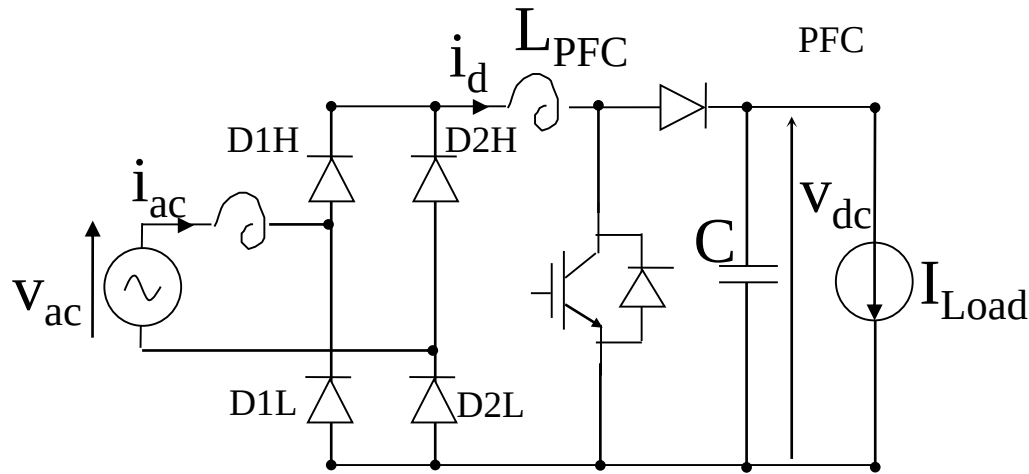
The period during which the grid is connected to the load is less than a semiperiod; therefore, given that the average value of i_d is equal to I_{load} , the peak of i_d is much higher than I_{load} and may produce distortion (notching) in the grid (EMI filter is required). The ac current is leading the voltage.



Increasing the capacitance of the dc-bus capacitor

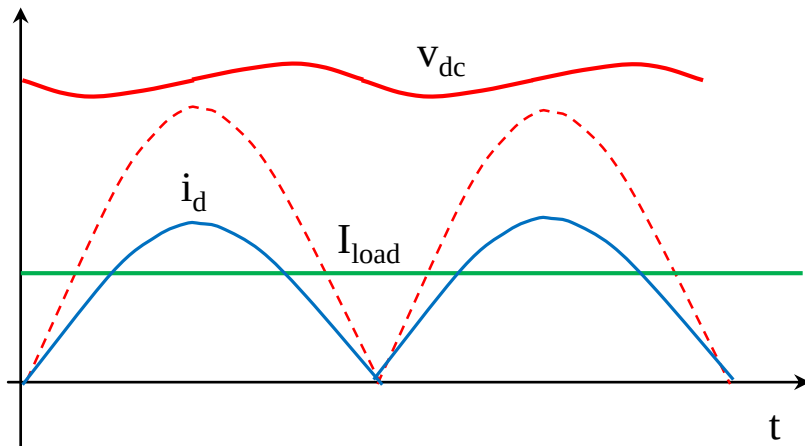


Increasing the capacitance, the period during which the grid is connected to the load is less than before; therefore, the peak of i_d is higher. The dc voltage ripple decreases but the distortion effect on the grid increases.

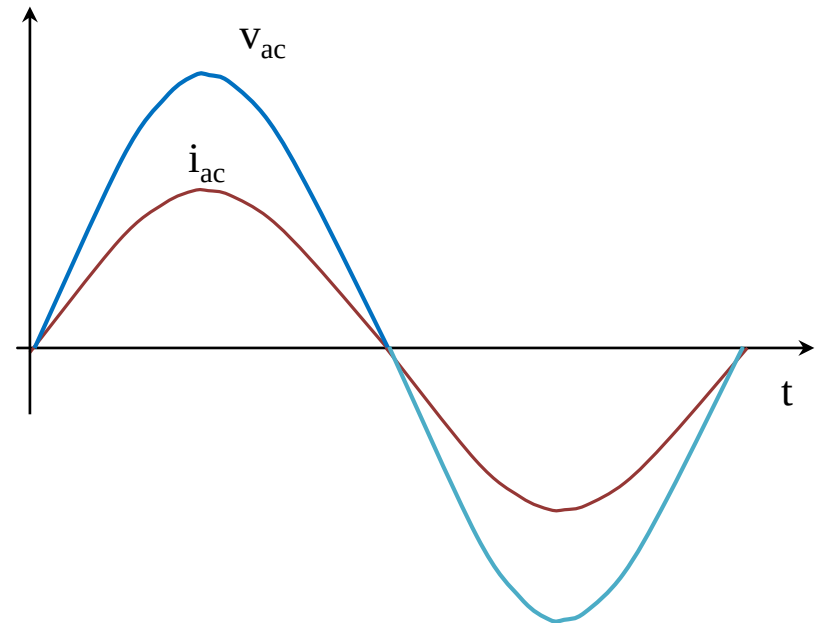


PFC (power factor correction) reduces the distortion (grid side), compensates the power factor and reduces the dc voltage ripple.

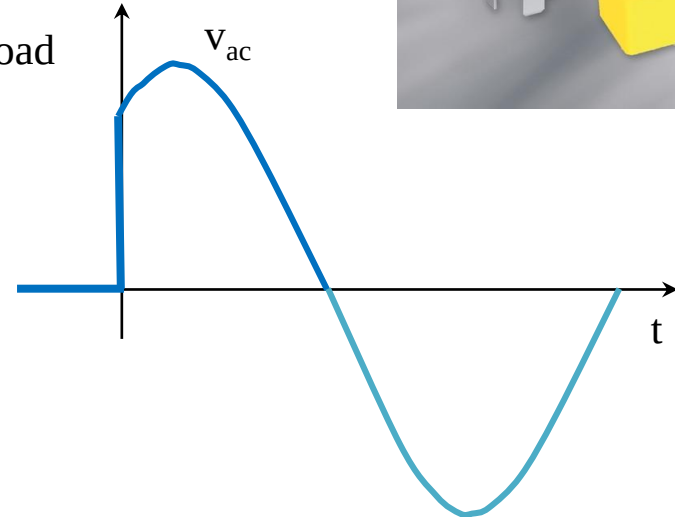
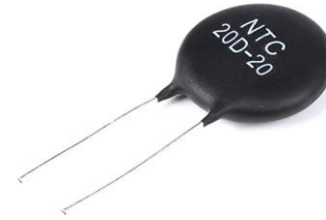
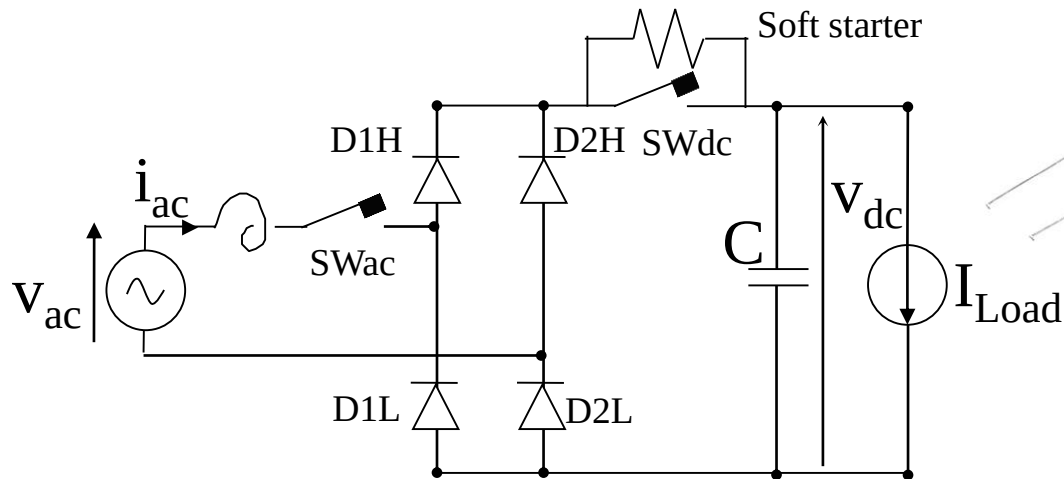
It needs more components.



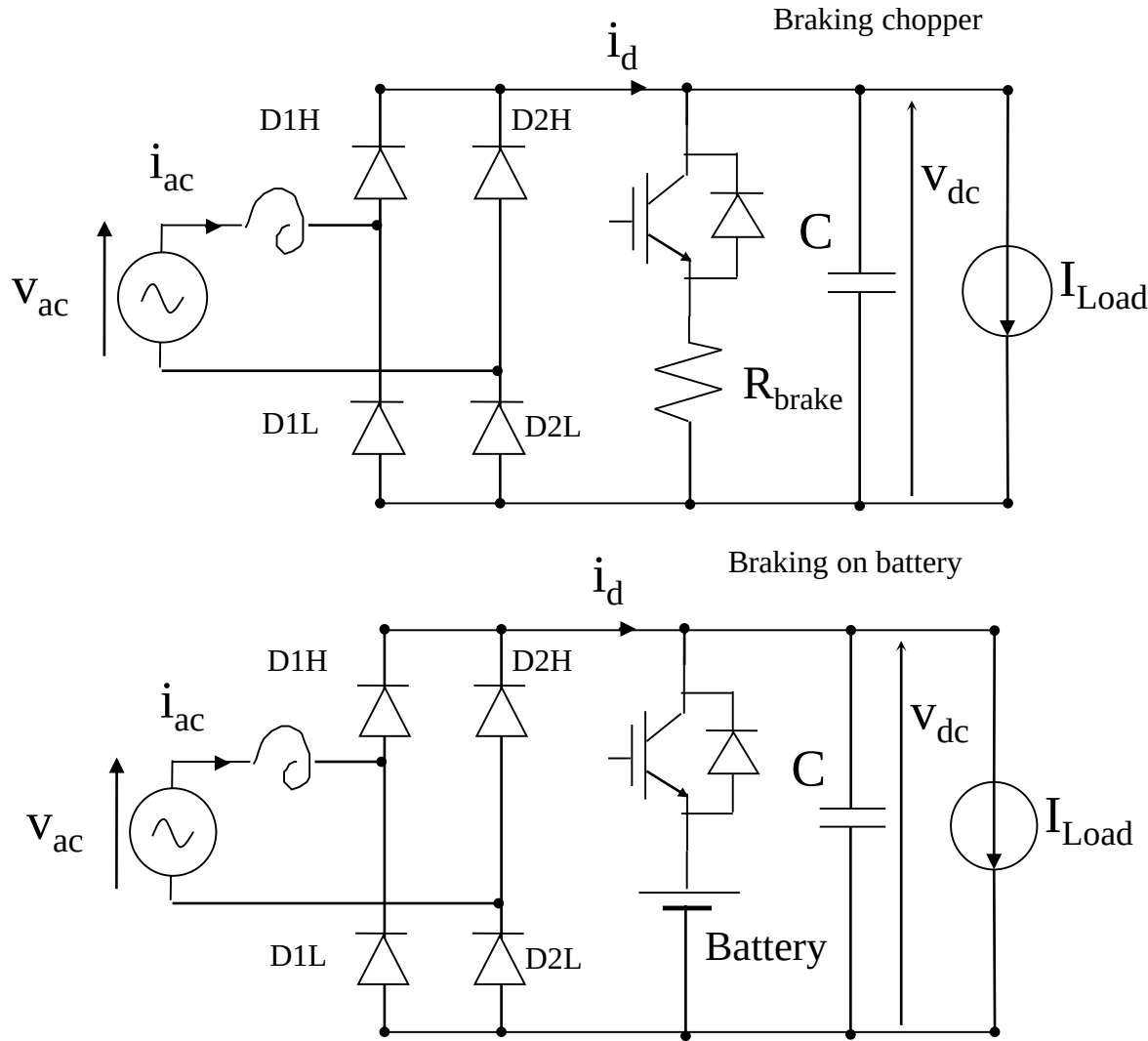
Diodes bridge and PFC require a soft-start device



Inrush current and Soft-start device



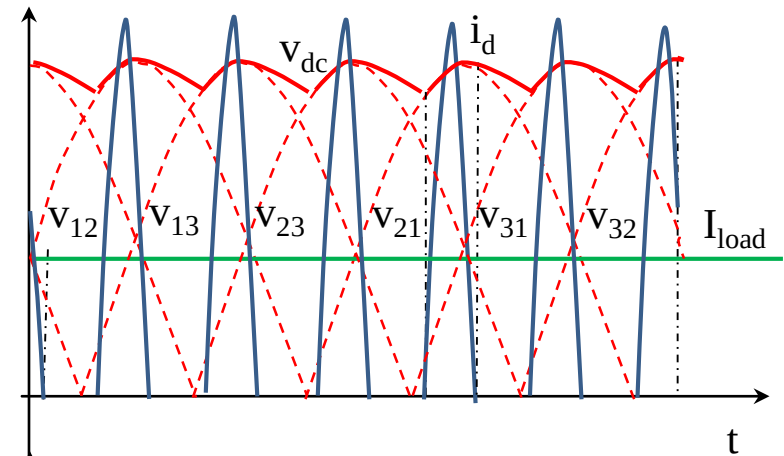
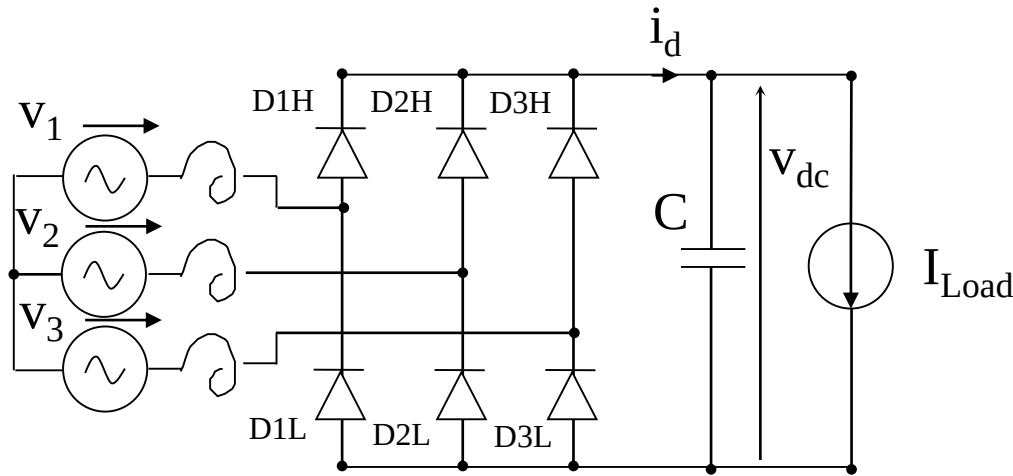
- Closing the switch on the ac side (SW_{ac}) without any pre-charging device, with the capacitor C discharged, the inrush current is very high, limited only by the impedance of the grid.
- In order to limit the inrush current, a soft-starter device has to be used: an NTC alone or a resistor with a contactor (SW_{dc}) as a bypass



During the braking phase the kinematic energy, stored in the mass/inertia of the mechanical load, are converted into electric energy. The instant power flow changes its sign, this means that I_{load} becomes negative.

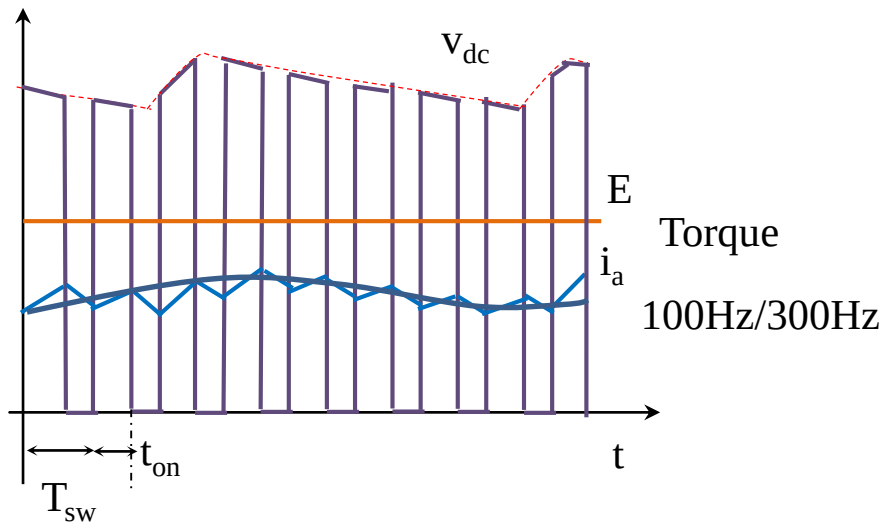
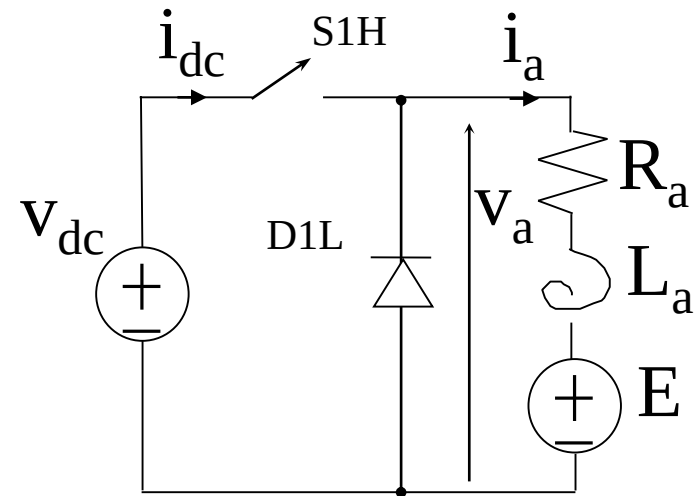
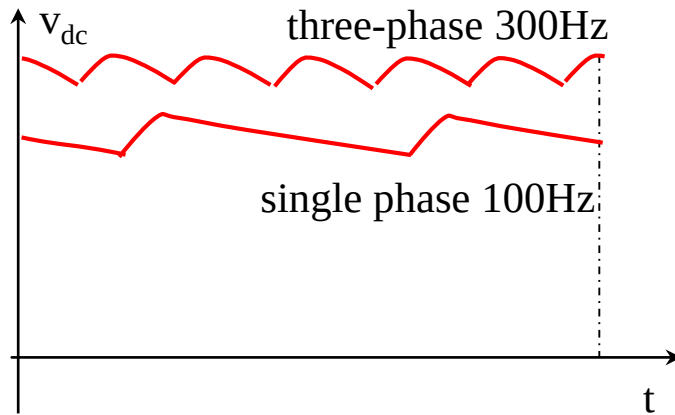
Diode bridge assures that $i_d \geq 0$, so the energy has to be dissipated or stored in a battery.

Three-phase diode bridge

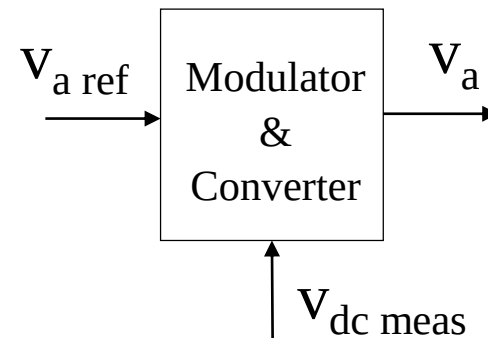


- Between N diodes with a common cathode, it conducts the one with the highest anode voltage
- Between N diodes with a common anode, it conducts the one with the lowest cathode voltage
- EMI filter

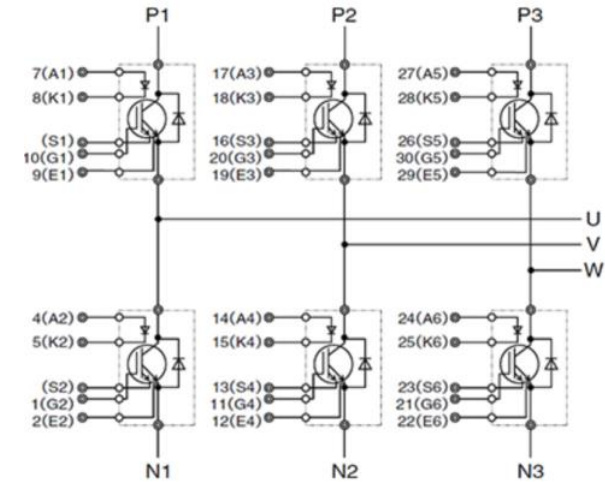
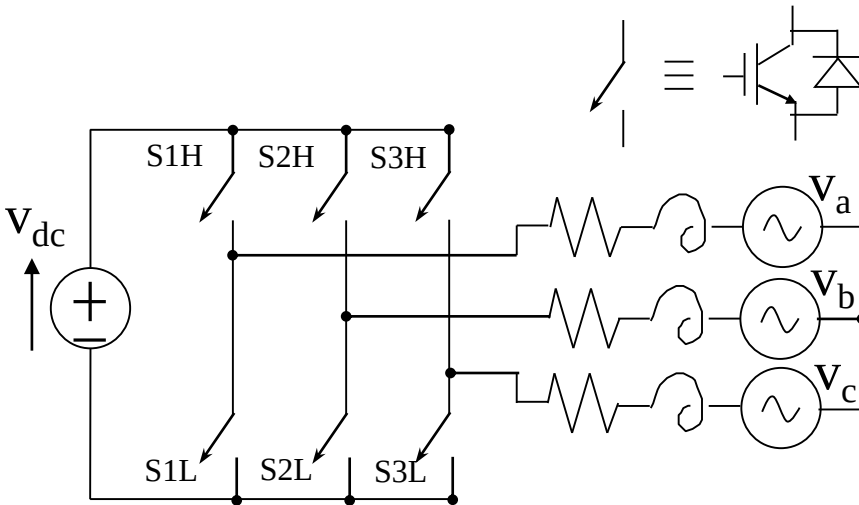
Compensation of the dc bus voltage oscillation



Power Supply



DC/AC: three-phase voltage source inverter (VSI)



Sinusoidal PWM with third harmonic injection (inscribed circle)

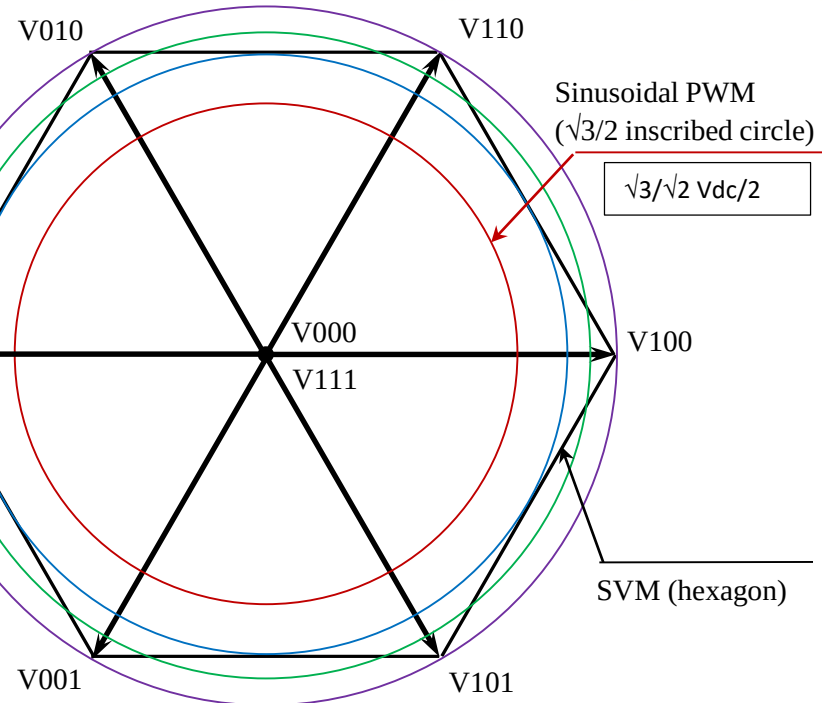
$$1/\sqrt{2} V_{dc}$$

Circumscribed circle

$$\sqrt{2}/\sqrt{3} V_{dc}$$

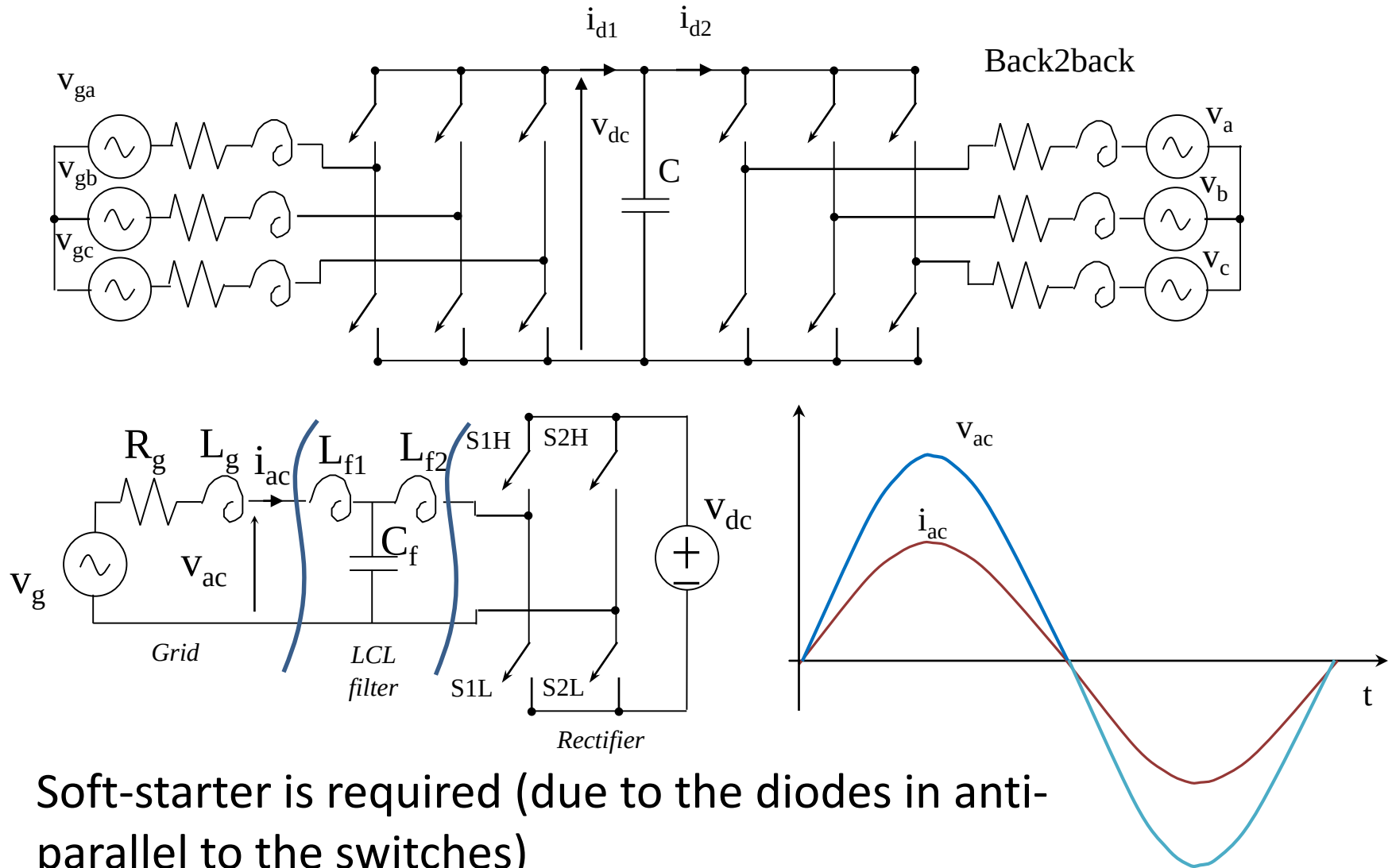
$$2/\pi \sqrt{3}/\sqrt{2} V_{dc}$$

Square wave first harmonic ($4/\pi \sqrt{3}/2$ inscribed circle)



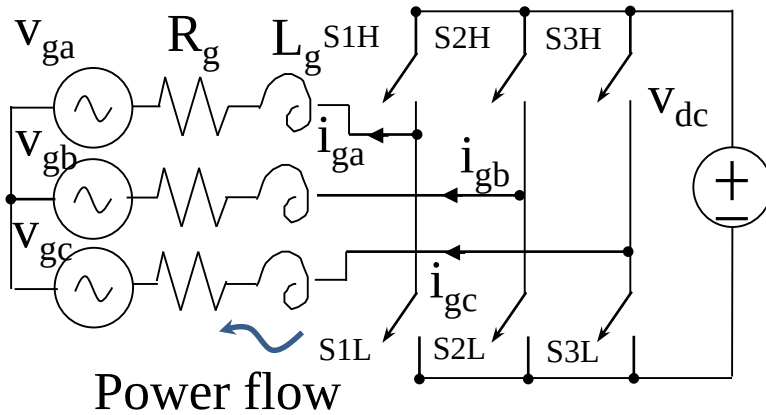
Back-to-back configuration

Called also “Regenerative rectifier” or “Active Front End” (AFE)



Soft-starter is required (due to the diodes in anti-parallel to the switches)

Regeneration



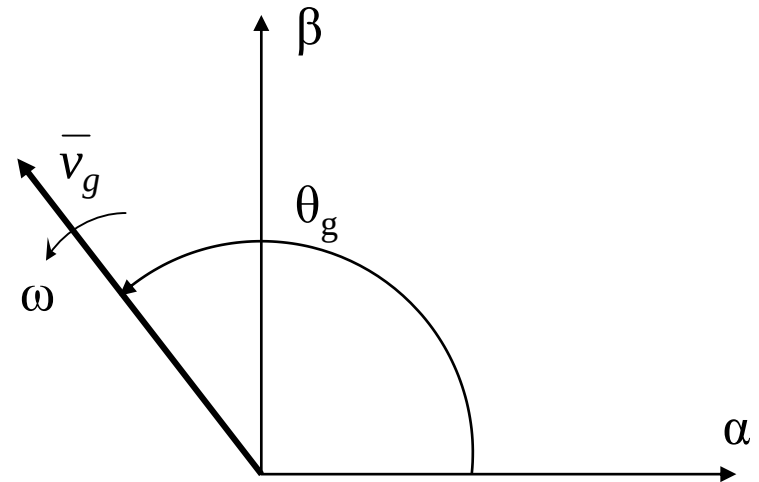
The grid has an equivalent circuit similar to that of an ac machine.

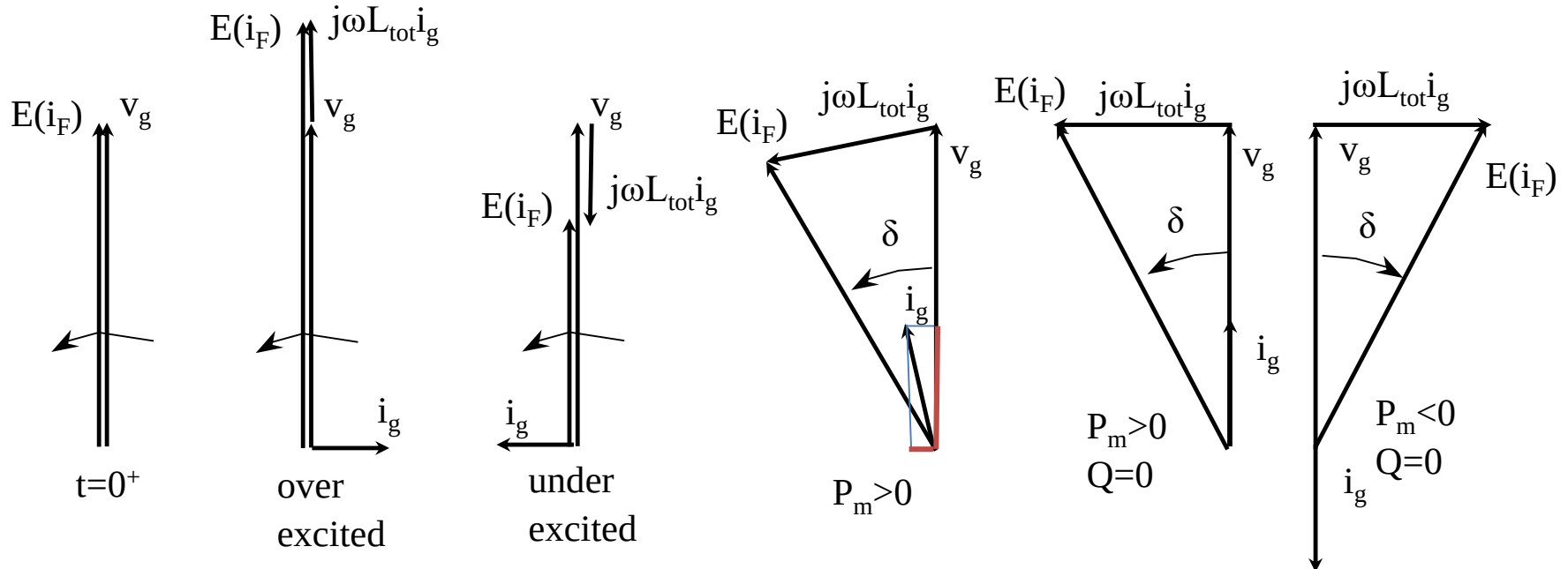
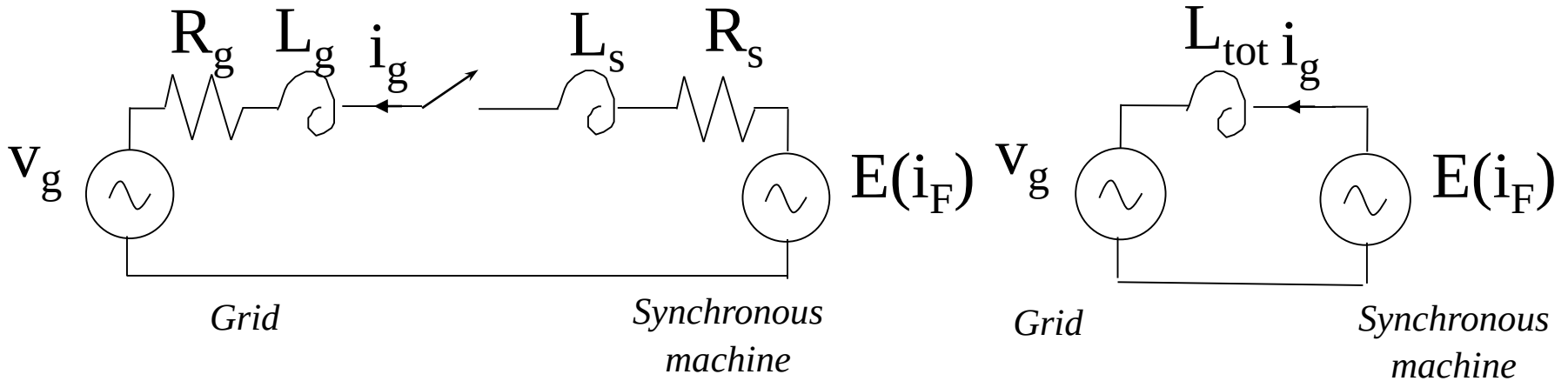
The main difference is that the voltage and the frequency are quite constant (in an ac machine the emf and the frequency are quite proportional to the mechanical speed)

The space phasor formula may be applied to the grid voltage v_g . In a stationary frame ($\alpha\beta$), it results ($v_{ga} = V_{ph\ max} \cos(\theta_g)$):

$$\overline{v_g} = V_g e^{j\theta_g}$$

Where V_g is the rms value of the line-line voltage, while θ_g the angle between the space phasor v_g and the α -axis





Control of power flow into the grid

Call v_i the voltage produced by the VSI (limited to the hexagon operating area). Choose a reference frame fixed with the grid voltage, where the imaginary axis (q) is aligned with the space phasor v_g . It results:

$$\bar{v}_i = R_g \bar{i}_g + L_g p \bar{i}_g + j \dot{\theta} L_g \bar{i}_g + \bar{v}_g$$

The components along d and q axes are:

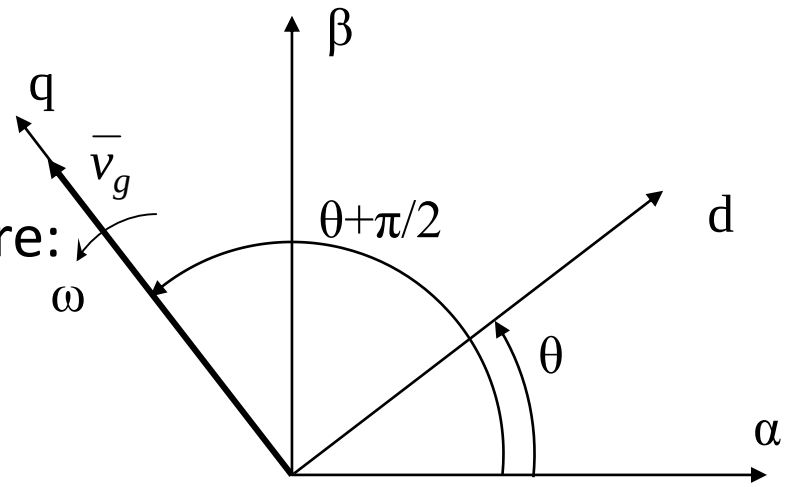
$$v_{id} = R_g i_{gd} + L_g p i_{gd} - \dot{\theta} L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + L_g p i_{gq} + \dot{\theta} L_g i_{gd} + v_g$$

The active and reactive power fed into the grid are:

$$P_g = \text{Re}(\bar{v}_g \underline{i}_g) = v_g i_{gq}$$

$$Q_g = \text{Im}(\bar{v}_g \underline{i}_g) = v_g i_{gd}$$



Control of active power fed into the grid

The derivative of the angle θ is the frequency ω .

Consider a steady state condition and a grid voltage constant in amplitude and frequency: $\overline{v_g} = V_g e^{j\omega t}$. Looking from the reference frame (dq), all the quantities are constant. Therefore:

$$\dot{\theta} = \omega$$

$$v_{id} = R_g i_{gd} - \omega L_g i_{gq}$$

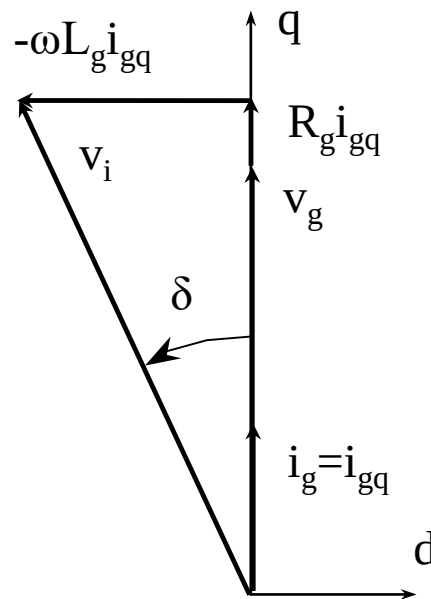
$$v_{iq} = R_g i_{gq} + \omega L_g i_{gd} + V_g$$

If the current i_{gd} (component along the d axis) is fixed to zero, the phasor diagram is:

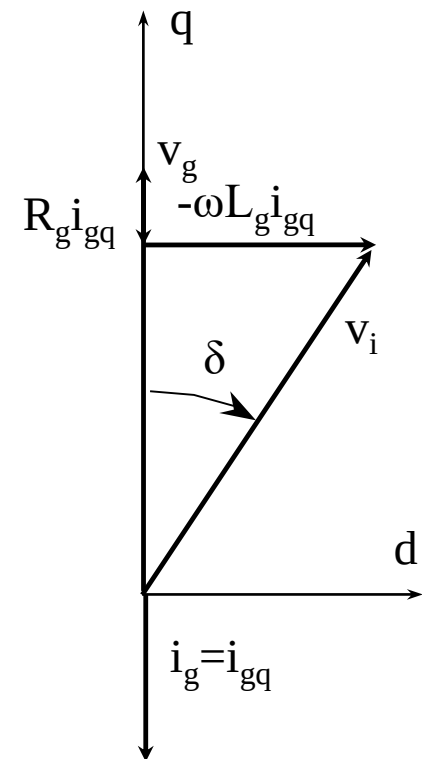
$$v_{id} = -\omega L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + V_g$$

Regenerative & $Q_g=0$



Rectifier & $Q_g=0$



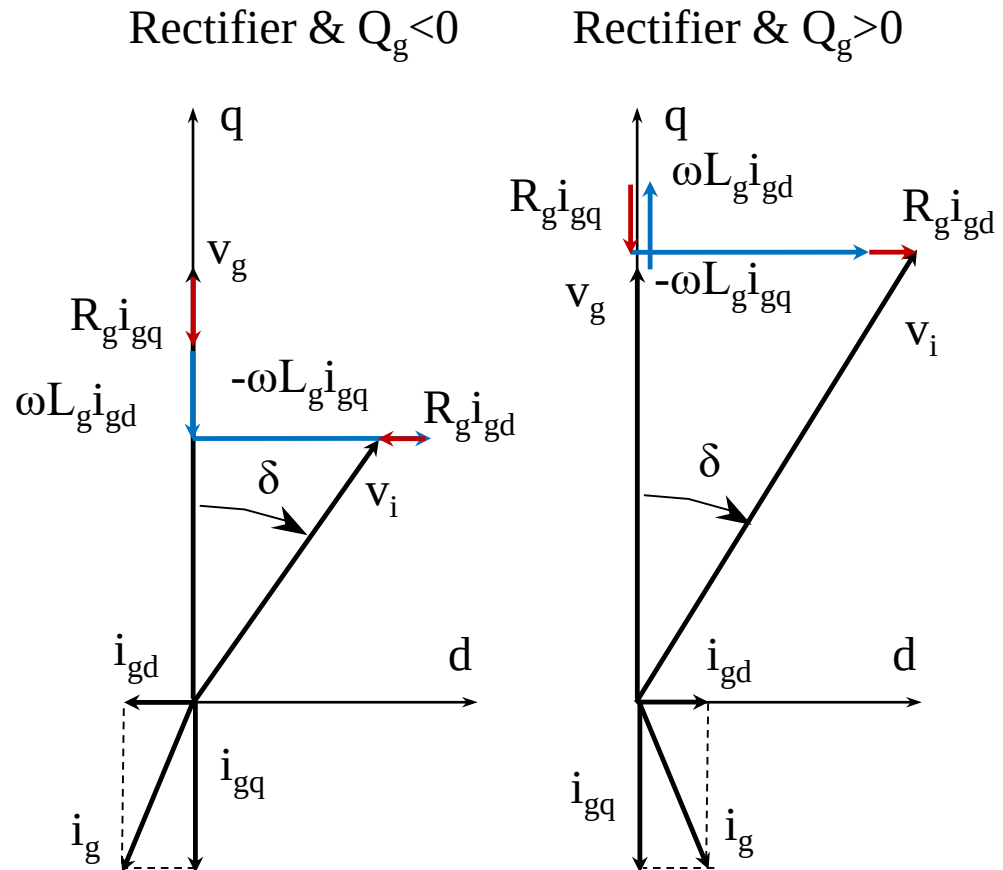
Control of active/reactive power fed into the grid

Controlling the current i_{gd} , it is possible to drive the reactive power: a positive value means that the inverter looks like a capacitor (and the grid like an inductor); vice versa for a negative value.

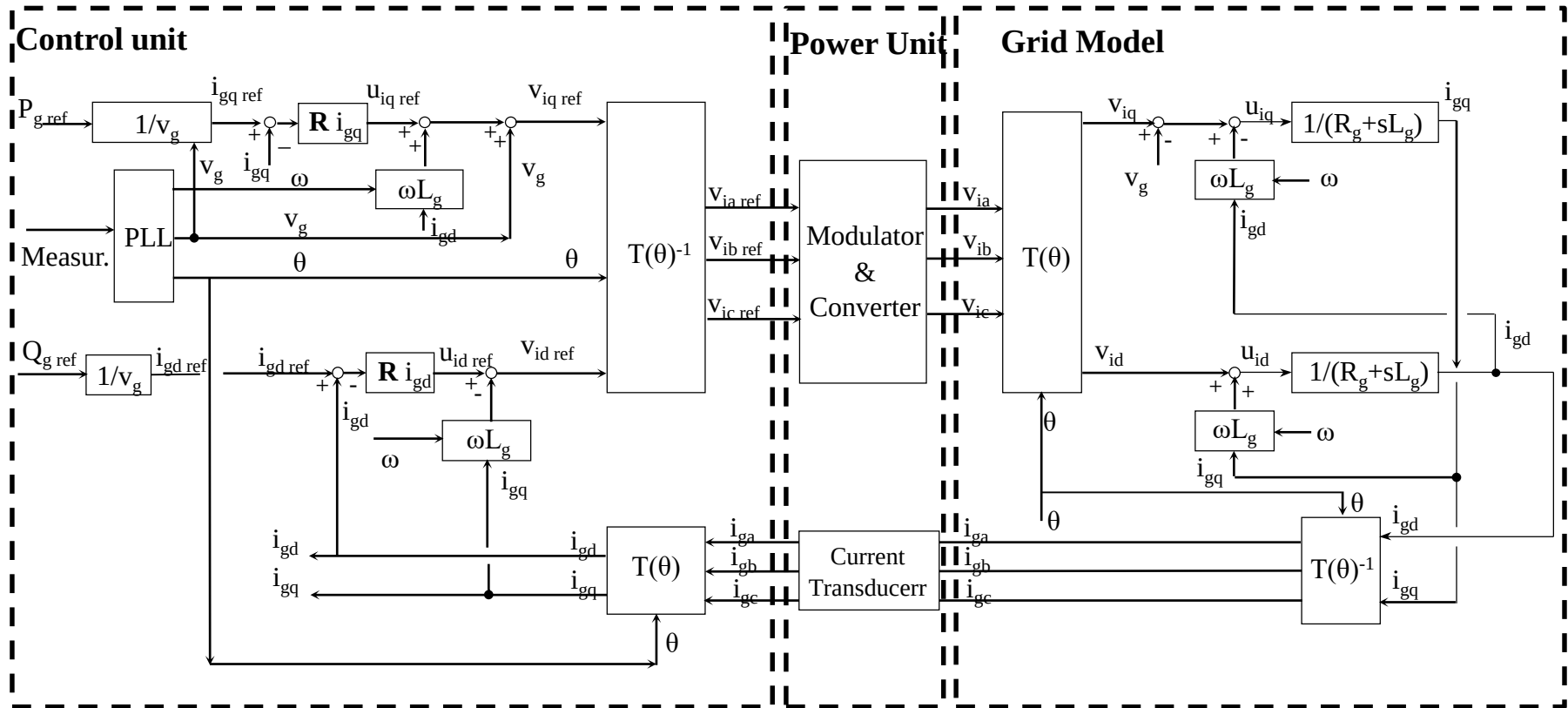
$$v_{id} = R_g i_{gd} - \omega L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + \omega L_g i_{gd} + V_g$$

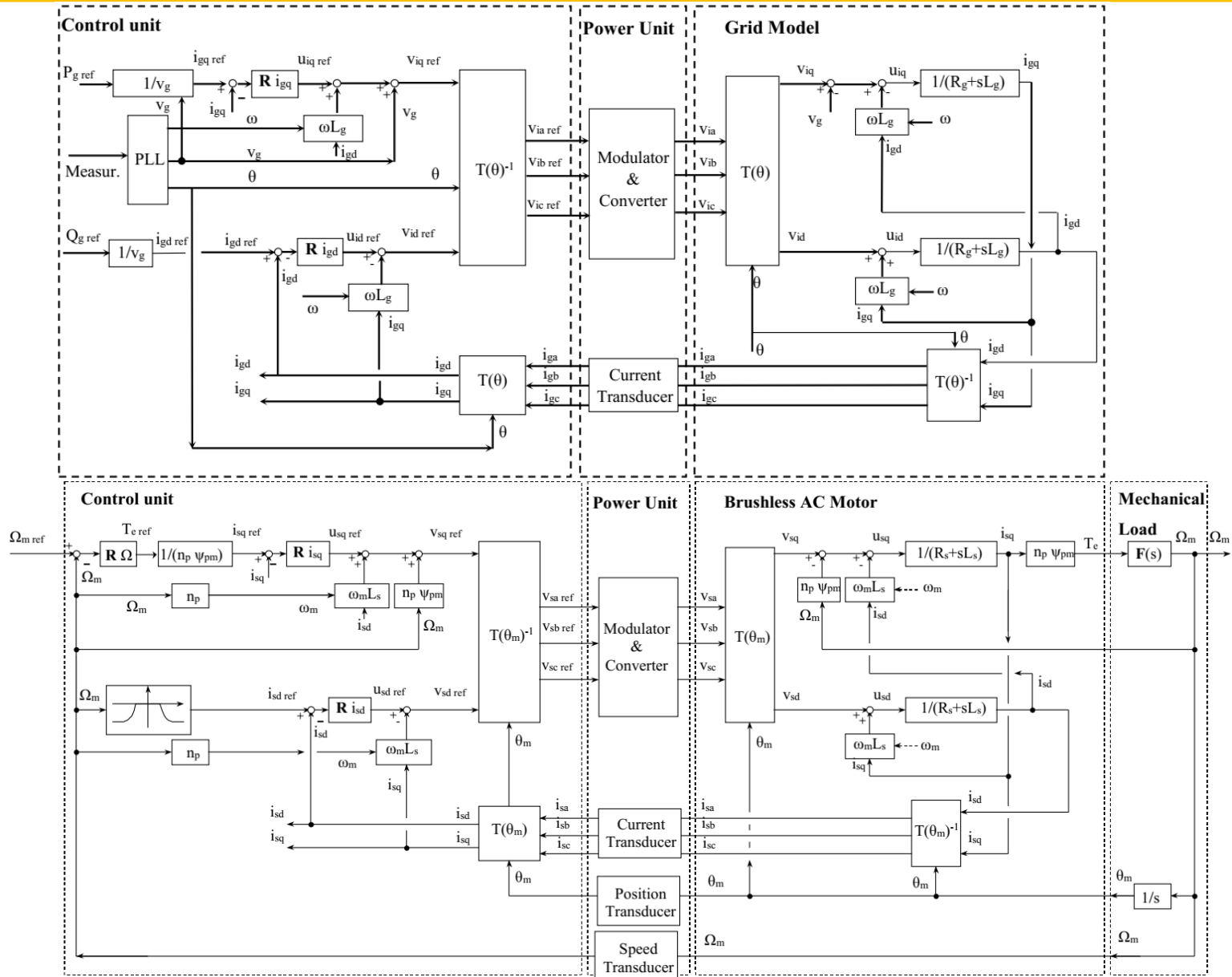
The figure represents the conditions when the active power is negative (operation as rectifier). Similar approach may be done for positive value of the active power (regeneration)



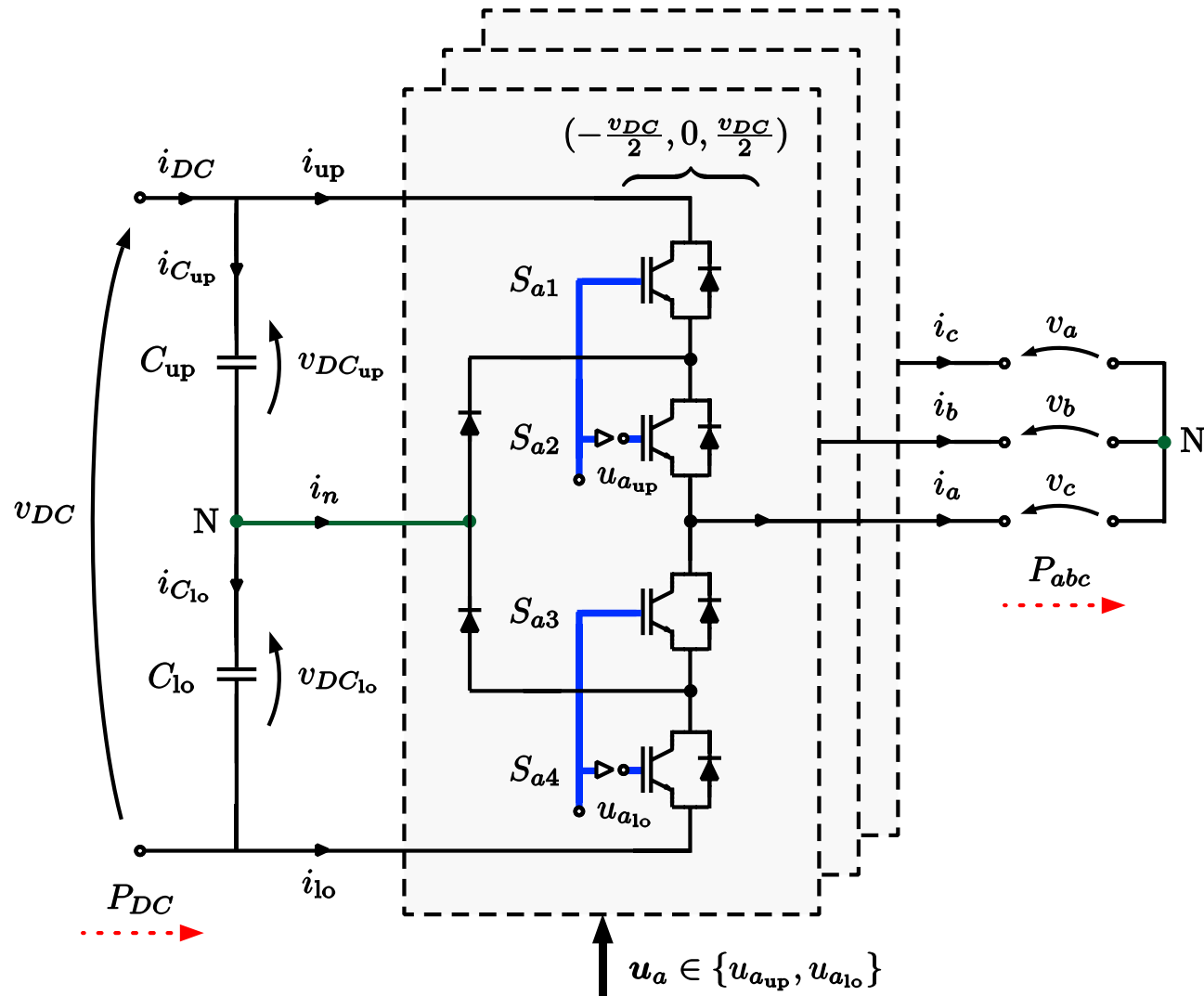
Control schemes



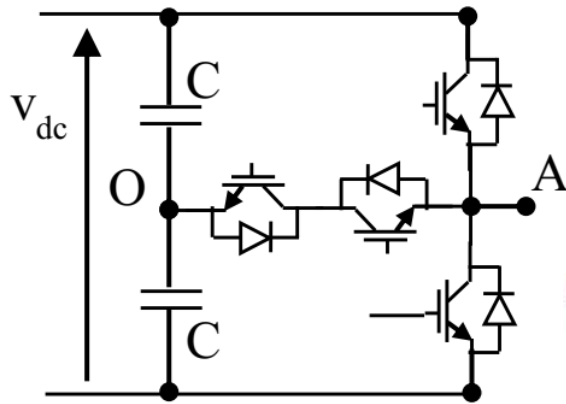
Control schemes: comparison with ac brushless



Multilevel converter: 3L-NPC

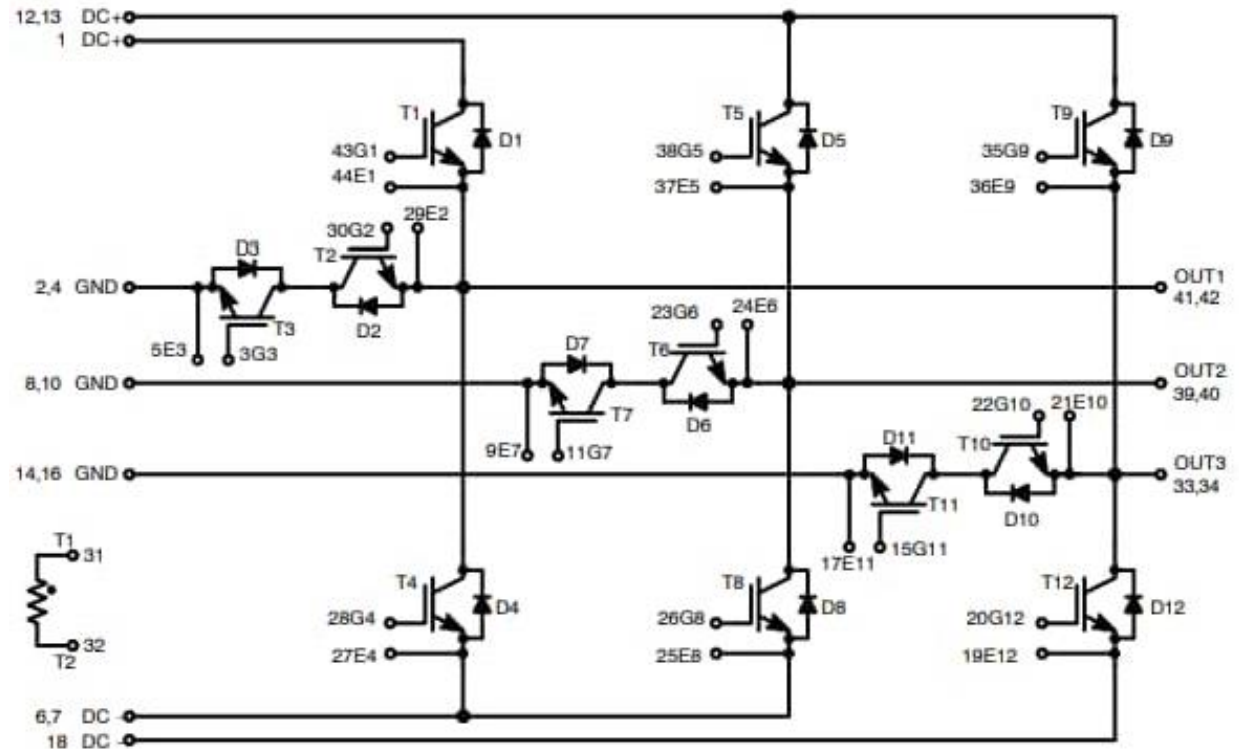


Multilevel converter: 3L-TNPCC



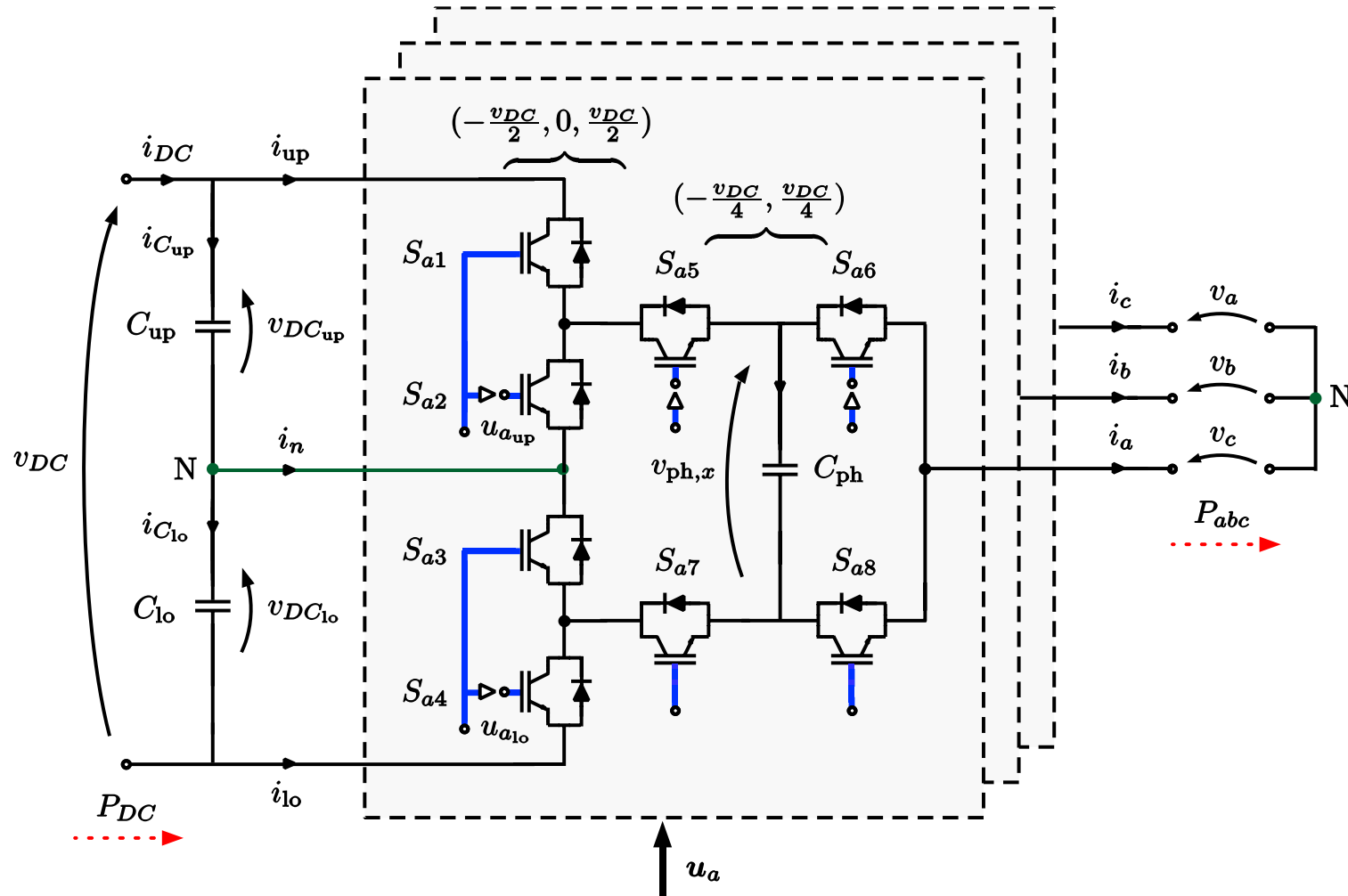
One leg

Three legs

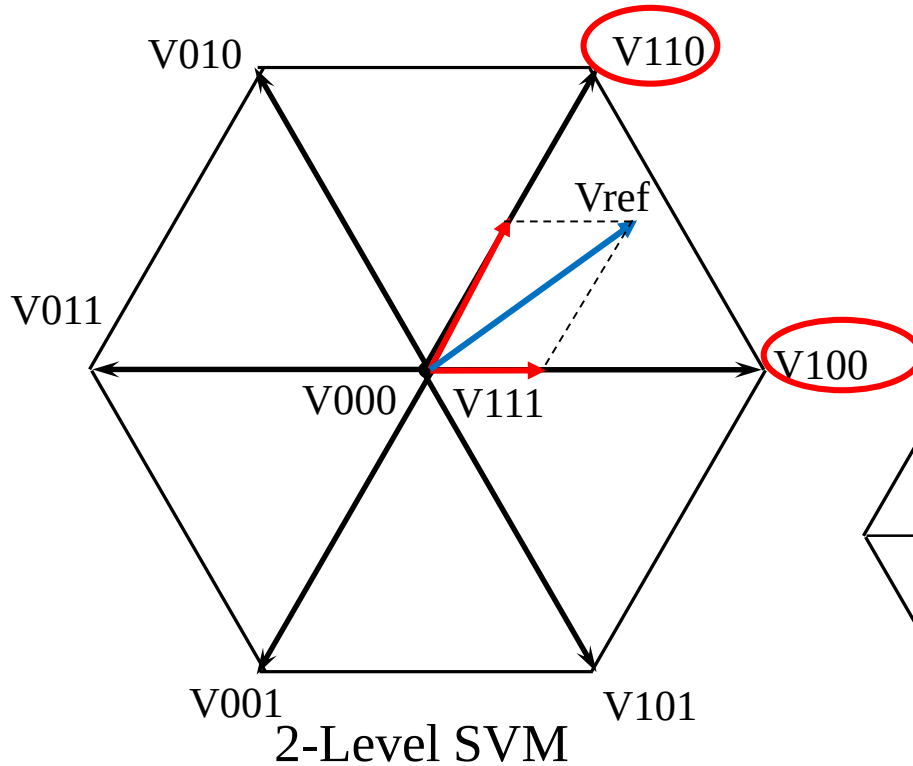


ON Semiconductor NXH40T120L3Q1 3-Phase TNPC Module

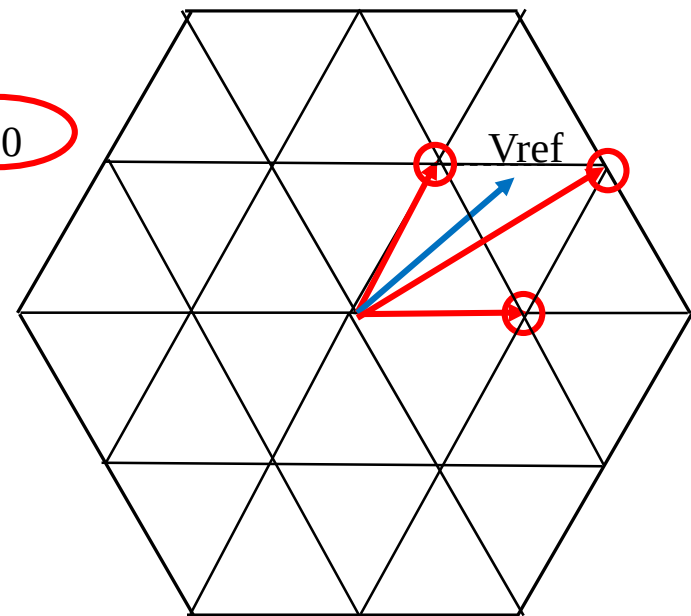
Multilevel converter: 5L-NPC



SVM for a 2L-VSC and 3L-NPC



3-Level SVM



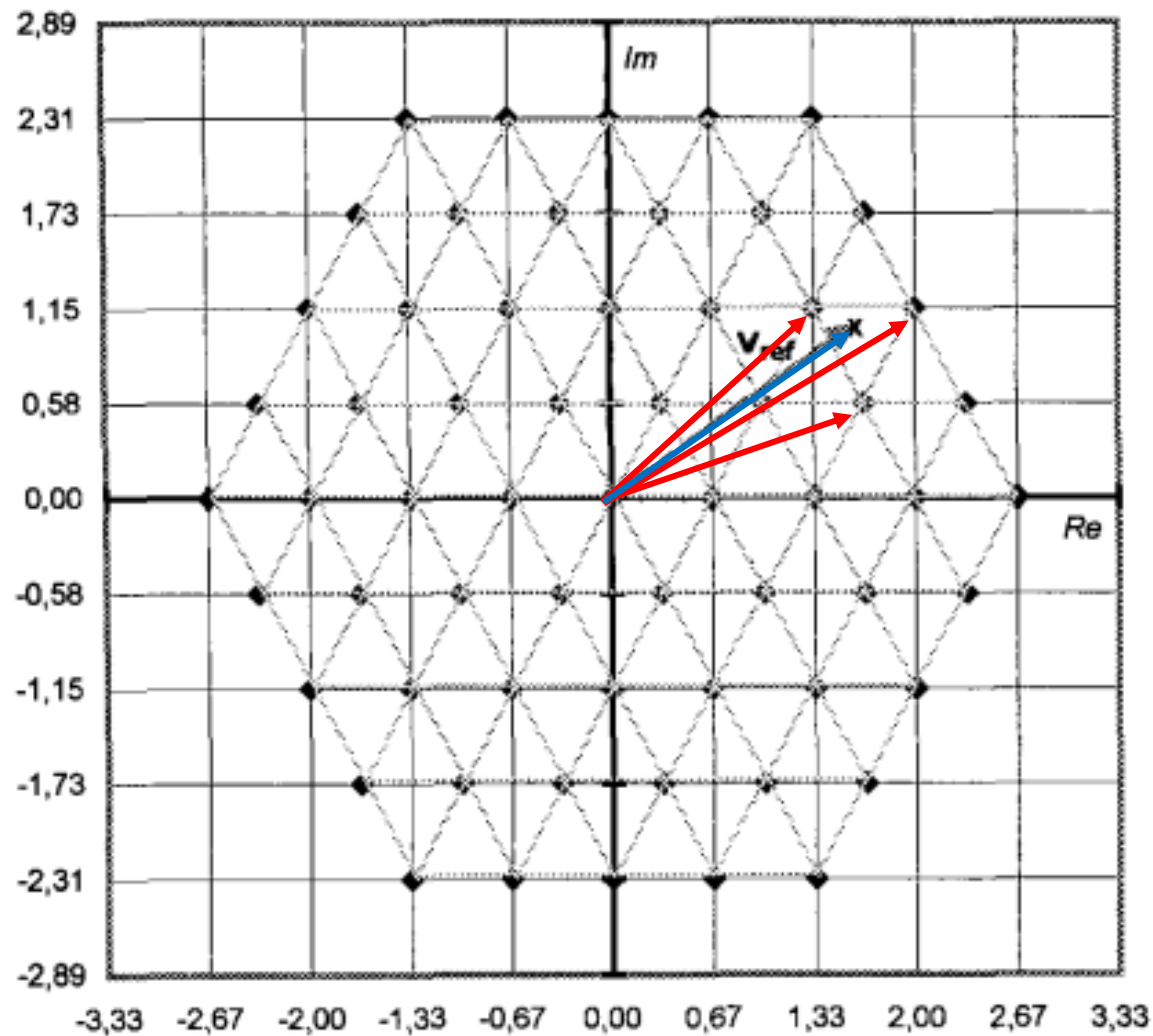


Fig. 3 Space-vector diagram of a 5-level converter

Multilevel converter: MMC

